Master’s Thesis

Investment Demand Shock and Business Cycle

投資需要ショックと景気循環

Graduate School of Economics

Department of Economics (UTIPE)

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Abstract of Master’s Thesis

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This paper asks whether an investment demand shock can serve as an alternative to the investment specific technology (IST) shock in accounting for the U.S. business cycle. The two shocks have a common observable footprint—both move investment—but a different economic origin: the IST shock raises the technology that converts consumption goods into investment goods, whereas the investment demand shock, following Nirei and Ragot (2025), traces back to idiosyncratic productivity shocks under lumpy investment and, in the aggregate, appears as a deviation of investment from the decision that households would otherwise have made. Because the shock perturbs investment after the endogenous investment decision is formed, I represent it in a representative-agent setting through a two-step solution: the first step solves the rational-expectations equilibrium with all frictions and shocks except the investment demand shock and pins down the endogenous investment decision; the second step holds that decision fixed, adds the exogenous shift in investment, and lets households re-optimize over consumption, labor, and other margins. I embed this device in a DSGE model with nominal rigidities, investment adjustment costs, and endogenous capital utilization, and estimate it on U.S. data from 1983:Q1 to 2002:Q4 using Bayesian methods.

In the baseline estimation the investment demand shock accounts for a sizable share of fluctuations. Under the unconditional forecast error variance decomposition it explains roughly 40% of the variance of output and 50% of investment, comparable in magnitude to the IST shock, and—unlike the IST shock—it generates an inflationary expansion. These figures should be read with caution rather than as a headline result. They are sensitive to how the variance is measured and to how the model is specified. At business-cycle frequencies (6–32 quarters) the same shock explains only about 10% of output and 16% of investment, because its effects largely die out within six quarters. Its contribution falls further, to about 4% of output and 14% of investment, in a larger Smets–Wouters-type model that adds habit formation, wage and price markup shocks, and consumption and wage data; and it changes again, to about 23% of output, when the shock is restricted to be transitory. The one specification in which its baseline importance survives is the model that uses the relative price of investment to discipline the IST shock and separates the marginal efficiency of investment shock, where it still accounts for roughly 43% of output and 49% of investment.

Beyond the variance shares, the estimates speak to the mechanism that distinguishes the two shocks. In the impulse responses the IST shock works mainly through intertemporal substitution—cheaper investment raises labor supply but depresses wages and consumption, leaving output nearly flat—while the investment demand shock operates through a wealth channel that shifts labor demand, raising hours, wages, and output together. This difference drives the inflation

result: writing inflation as the present value of marginal cost, the investment demand shock raises both the wage and the capital-rental components and so is inflationary across the posterior draws and parameter sweeps, whereas the sign of the IST shock's inflation response is itself fragile, flipping with the shock persistence and the capital-utilization elasticity. The positive consumption–investment comovement emphasized by Nirei and Ragot (2025) appears only under specific combinations of the Phillips-curve slope, the Taylor-rule inflation coefficient, and wage rigidity, when the model is sticky enough to mute the short-run substitution effect. A historical decomposition shows the investment demand shock to be the more volatile and procyclical of the two over the sample, moving closely with output and inflation.

A few limitations qualify these findings. Identification is indirect: no series pins down the investment demand shock, so in the baseline its separation from the IST shock rests only on their differing effects on inflation and consumption, which the high-dimensional parameter space lets me check only along one-parameter sweeps rather than exhaustively. The horse race against the marginal efficiency of investment shock sharpens this concern—because several shocks can be written into the same capital law of motion, an estimated shock may capture little more than that equation's residual, whatever name it is given. The model also fits the data imperfectly: the variance of hours worked is substantially overstated, traceable to an excessively volatile consumption process, and several posteriors settle at near-extreme values.

Taken together, the results suggest that the investment demand shock can be quantitatively important and is a plausible competitor to the IST shock, but that this conclusion is fragile: it depends materially on the measurement of variance, the richness of the model, the assumed shock process, and an identification strategy that the data only partially support. The broader lesson is methodological—structural estimation can surface economically meaningful mechanisms, but the quantitative weight it assigns to a weakly identified shock should be treated as conditional on specification rather than as a robust fact.

1 Introduction

Investment has been seen as an important part of explaining the business cycle for a long time. It remains the most volatile variable in the macroeconomy. Over the past few decades, the role of investment in the business cycle has been continuously discussed, with new model environments being proposed (Greenwood et al., 1988, 1997; Fisher, 2006; Justiniano et al., 2010; Winberry, 2021).

Early research has drawn attention to the investment shock because of the obvious trend in the data on the relative price between investment goods and consumption goods, which implies another trend term in the investment specific technology besides the neutral technology (Greenwood et al., 1988). After that, quantitative results have been established by Greenwood et al. (1997) using a calibrated DSGE model, and by Fisher (2006) using a structural VAR model. Both of them claimed that the investment shock accounts for about half of the forecast error of the output and the working hours.

A new wave of research about the investment shock is stimulated by Smets and Wouters (2007), who estimated a more frictional model based on Christiano et al. (2005) using Bayesian estimation. Using the same approach, Justiniano et al. (2010, 2011) argued again that the investment shocks are the main source of the business cycle. While the investment shock shows strong effects on the macroeconomic variables, there are other candidates being proposed to account for the business cycle. Christiano et al. (2014) and Kamber et al. (2015) argued that investment shock is less important once the model includes financial friction and shocks in the financial market.

Though I use the name of investment shock in the previous paragraphs, there are various kinds of specifications being discussed. Among them, the most classic one is the investment specific technology (IST) shock, which increases the technology level to produce investment goods from consumption goods (Greenwood et al., 1997; Fisher, 2006; Justiniano et al., 2010). Besides, Justiniano et al. (2011); Kamber et al. (2015) distinguished the IST shock from the marginal efficiency of investment (MEI) shock, which represents the technology level to produce capital goods from investment goods. However, the difference between these two shocks relies on the detailed modeling of the capital production process. If the capital production is modelled by one capital law of motion equation, then those two shocks cannot be distinguished in their effect, and thus I make no clarification between them except for Section 6.3. In addition, many works have discussed the IST news shock, in the sense that the shock is known by agents now but will affect the economy several periods later (Schmitt-Grohé and Uribe, 2012; Ben Zeev and Khan, 2015; Liao and Chen, 2023). However, this kind of shock is beyond the scope of this paper.

Although the IST shock has been considered an important driving force of the business cycle, it fails to generate the comovement between consumption, investment and inflation. To solve the comovement problem, Nieri and Ragot (2025) proposed an investment demand shock based on a model featuring firms' heterogeneity and the lumpy investment assumption [1]. With the calibrated model, the investment demand shock generates the comovement between consumption and investment. While the shock is traced back to idiosyncratic productivity shocks, at the aggregate level it appears as a deviation of investment from the original investment decision made in a shock-free environment. Because of this, the investment demand shock can also be represented in a DSGE model without heterogeneity, and be estimated using the same approach as Smets and Wouters (2007).

However, since this shock requires fixing the endogenous investment decision, the solution method of the rational expectation equilibrium needs to be separated into two steps. In the first step, the model includes all shocks and frictions except for the investment demand shock, from where the endogenous investment decision is solved. In the second step, the investment is decided by both the endogenous investment decision and the exogenous investment demand shock. It is worth pointing out that the endogenous investment decision cannot be changed in the second step, and thus investment demand is exogenously increased by the shock. Although the investment decision is fixed, households can still adjust their decisions on other terms, such as consumption and labor input, and thus the second step solves another equilibrium. This equilibrium, finally, will be used to compute the likelihood.

The identification strategy is not direct—it is hard to directly identify the investment demand shock from data, as the shock originates from idiosyncratic productivity shocks. Moreover, though we can use the relative price of investment goods as a proxy for the IST shock in some sense, there is still a concern that we might attribute all residuals to the investment demand shock. Therefore, I postpone this exercise to a later section and rely on the theoretical difference—whether the shock generates comovement between consumption, investment and inflation—to identify the two shocks in the baseline result. However, due to the high-dimensional parameter space, the comovement difference is hard to test across all combinations of parameter values. Therefore, I acknowledge the shortcoming of such an identification strategy.

Section 2 introduces the baseline model. Section 3 presents the solution method and incorporates the investment demand shock. Section 4 describes the Bayesian estimation, and Section 5 reports the benchmark results. Section 6 provides robustness checks, and

[1] Due to lumpy investment, there is an inaction region of current capital in which firms will not invest. This region is determined by the production cost and the expected demand faced by the firm.

Suppose there are some high-productivity firms endowed with an idiosyncratic productivity shock. First, the production cost of these firms will decrease; thus other firms that use their products as intermediate materials enjoy a lower cost as well, and may increase production and investment. Second, those high-productivity firms may increase production themselves, which pushes up aggregate demand; thus other firms may enjoy higher demand for their products, and increase their investment as well.

In summary, through the supply channel and the demand channel, the idiosyncratic productivity shock propagates to the whole economy. For the propagation effect to be large enough, the friction of lumpy investment plays an important role. Because of this friction, firms nearly always invest a little more than their optimal investment, which can amplify an idiosyncratic shock into a macro-level shock.

Section 7 concludes.

2 Baseline Model

The baseline model is based on Herbst and Schorfheide (2015) and Smets and Wouters (2007). I selected some model settings to maintain the fit to the data, including price and wage rigidity, investment adjustment cost, and endogenous capital utilization ratio. However, some settings are left out in the baseline model to keep the solution clean, such as habit formation and price indexation.

I first specify the model without the investment demand shock, and then add the shock in Section 3.

2.1 Household

There is a continuum of households indexed on $[0, 1]$. For simplicity of notation, the index of the household is omitted in the following equations. The representative household owns the firms and solves the dynamic optimal problem over consumption c_t, hours worked ℓ_t, nominal bonds B_t, investment i_t, capital k_t, and capital utilization u_t.

$$\max_{c,b,i,k,l,u} \mathbb{E}_t \sum_{k=0}^{\infty} \beta^k \left(\frac{(c_{t+k}/\gamma^{t+k})^{1-\sigma} - 1}{1-\sigma} - \chi \frac{\ell_{t+k}^{\psi+1}(j)}{\psi + 1} \right) \tag{1}$$

In the objective function, γ is the growth rate of the labor-augmented technology, which is used to generate a trend in the state variables. Usually, if we use detrended data, there is no need to add this term. However, the process of detrending itself includes some operations on the data, which are not explained in the model. To maintain the integration of the estimation process, I use γ to fit the non-detrended data.

Following Herbst and Schorfheide (2015), I assume the representative household derives utility from effective consumption, defined as absolute consumption scaled by the trend in labor-augmenting technology. This formulation mathematically guarantees the existence of a balanced growth path and the convenience of representing the marginal utility, as we can directly define the detrended consumption $\tilde{c}_{t+k} = c_{t+k}/\gamma^{t+k}$ within it.

The budget constraint appears in the following equation:

$$c_t + i_t + \frac{b_{t+1}}{R_{t+1}P_t} = r_t^k k_{t-1} u_t - a(u_t) k_{t-1} + \frac{b_t}{P_t} + w_t \ell_t - \frac{\tau_t}{P_t} + \frac{d_t}{P_t}. \tag{2}$$

where b_t and R_t denote nominal bonds and the nominal interest rate. τ_t is the nominal tax, and d_t is the nominal dividend from the firms they own. $a(u_t)k_{t-1}$ is the adjustment cost of capital utilization. $a(\cdot)$ is a function defined on $[0, 1]$, satisfying $a(1) = 0$ and $a''(1) > 0$. Endogenous capital utilization allows capital to be adjusted when facing shocks, even though capital is still a predetermined variable, and the adjustment cost of capital utilization smooths the adjustment process.

The law of motion of capital adds several additional terms compared with the standard one.

$$k_{t+1} = (1-\delta)k_t + \mu_{s,t}\left[1 - S\left(\frac{i_t}{i_{t-1}}\right)\right] i_t \tag{3}$$

The first one is the investment specific technology shock $\mu_{s,t}$, which can be assumed to follow a stationary AR(1) process without incorporating the relative price of investment.

$$\ln \mu_{s,t} = (1-\rho_i)\ln \mu_s + \rho_s \ln \mu_{s,t-1} + \varepsilon_{s,t}, \ \ \varepsilon_{s,t} \sim \text{i.i.d. } \mathcal{N}(0, \sigma_s^2) \tag{4}$$

The second one is the investment adjustment cost $S\left(\frac{i_t}{i_{t-1}}\right)$. I use the function form $S(x) = \frac{\phi_i}{2}(x-\gamma)^2$, indicating that if the growth of the investment is above or below the balanced path growth rate, then an additional adjustment cost should be charged. The setting is important because the investment would otherwise be very sensitive to the shocks, which fails to match the comovement in the data (Winberry, 2021) and causes unrealistic impulse response functions.

From the household's optimal problem, we obtain the following first-order conditions. Let Q_t denote the ratio of the Lagrange multiplier on the law of motion of capital to the Lagrange multiplier on the budget constraint, which captures the relative marginal utility between consumption and investment. Q_t has the following relation to Tobin's Q: Tobin's $Q = Q_t \mu_{s,t}$. Define gross inflation as $\pi_{t+1} \equiv P_{t+1}/P_t$.

$$\Lambda_{t,t+1} \equiv \frac{\beta}{\gamma}\left(\frac{c_{t+1}/\gamma^{t+1}}{c_t/\gamma^t}\right)^{-\sigma} \tag{5}$$

$$\begin{aligned} 1 = Q_t\, \mu_{s,t} & \left(1 - S\left(\frac{i_t}{i_{t-1}}\right) - S'\left(\frac{i_t}{i_{t-1}}\right)\frac{i_t}{i_{t-1}}\right) \\ & + \mathbb{E}_t\left[\Lambda_{t,t+1}\, Q_{t+1}\, \mu_{s,t+1}\, S'\left(\frac{i_{t+1}}{i_t}\right)\left(\frac{i_{t+1}}{i_t}\right)^2\right], \end{aligned} \tag{6}$$

$$Q_t = \mathbb{E}_t\left[\Lambda_{t,t+1}\left(r^k_{t+1}u_{t+1} - a(u_{t+1}) + (1-\delta)Q_{t+1}\right)\right], \tag{7}$$

$$1 = \mathbb{E}_t\left[\Lambda_{t,t+1}\frac{R_{t+1}}{\pi_{t+1}}\right], \tag{8}$$

$$w_t = \chi\, \ell_t^{\psi}\, \gamma^t \left(\frac{c_t}{\gamma^t}\right)^{\sigma}, \tag{9}$$

$$r^k_t = a'(u_t). \tag{10}$$

2.2 Final Goods Producer

This part of model is standard, where perfectly competitive firms produce the final goods Y_t using a CES aggregator. They solve the optimal problem,

$$\begin{aligned} \max_{Y_t, y_t(j)} \; & P_t Y_t - \int_0^1 p_t(j) y_t(j)\, dj \\ \text{s.t. } Y_t &= \left(\int_0^1 y_t(j)^{\nu-1}\, dj \right)^{\frac{1}{\nu-1}}. \end{aligned} \tag{11}$$

Therefore, from the first-order condition, the demand for each intermediate good follows:

$$y_t(j) = \left(\frac{p_t(j)}{P_t} \right)^{-\nu} Y_t, \tag{12}$$

where P_t follows

$$P_t = \left(\int_0^1 p(j)^{\frac{\nu-1}{\nu}}\, dj \right)^{\frac{\nu}{\nu-1}}. \tag{13}$$

Alternatively, we could define the price markup shock by replacing the fixed parameter ν by a time-varying random variable.

2.3 Intermediate Goods Producer

Each intermediate goods producer has to solve both the cost minimization problem and the dynamic pricing problem.

The cost minimization problem is

$$\min_{k_t^s(j), n_t(j)} \; r_t^k k_t^s(j) + w_t n_t(j) \tag{14}$$

subject to their production technology,

$$y_t(j) = \mu_{a,t} k_t^s(j)^{\alpha} \left(\gamma^t n_t(j) \right)^{1-\alpha}, \tag{15}$$

where $\mu_{a,t}$ is the aggregate productivity technology, following

$$\ln \mu_{a,t} = (1-\rho_a) \ln \mu_a + \rho_a \ln \mu_{a,t-1} + \varepsilon_{a,t}, \;\; \varepsilon_{a,t} \sim \text{i.i.d. } \mathcal{N}(0, \sigma_a^2). \tag{16}$$

γ^t ($\gamma > 1$) is the labor-augmented technology, which helps derive the balanced growth path. In addition, K_t^s means the capital service, which will be influenced by the capital utilization u_t under general equilibrium conditions.

The cost minimization solves the cost function. However, as the production function has constant returns to scale, the marginal cost is solved as well:

$$mc_t(j) = \frac{1}{\alpha^{\alpha}(1-\alpha)^{1-\alpha}\gamma^{t(1-\alpha)}} \frac{w_t^{1-\alpha}(k_t^s)^{\alpha}}{\mu_{a,t}}. \tag{17}$$

The Calvo pricing problem is described below.

$$\begin{aligned} &\max_{p_t(j)} \mathbb{E}_t \sum_{k=0}^{\infty} \theta_p^k \Lambda_{t+k,t} \left(p_t(j) y_{t+k}(j) - P_{t+k} mc_{t+k}(j)\right) \\ &\text{s.t. } y_{t+k}(j) = \left(\frac{p_t(j)}{P_{t+k}}\right)^{-\nu}, \end{aligned} \tag{18}$$

where θ_p is the Calvo probability of not being allowed to reoptimize one's price, and $\Lambda_{t+k,t}$ is the stochastic discount factor between period $t+k$ and period t defined in equation (5).

I drop the price indexation, which allows firms to move their prices according to the inflation in the last period even though they are not allowed to reoptimize their prices, because Smets and Wouters (2007) shows the model has a higher posterior likelihood without this setting.

2.4 Wage Setting

Instead of modeling a detailed intermediate labor sector, the baseline model uses a simplified wage setting equation to introduce the sticky wage, following Nieri and Ragot (2025).

$$w_t = \left(\frac{L_t^{\psi}}{C_t^{-\sigma}}\right)^{\theta_w} w^{1-\theta_w} \tag{19}$$

where w is the steady-state real wage. Similar to the Calvo pricing problem, the equation roughly means that only a $\theta_w \in (0,1)$ part of wage contracts can be reset to match the current marginal rate of substitution between hours worked and consumption. Meanwhile, the rest of the $1-\theta_w$ part can only keep the steady-state real wage.

Comparing with the standard sticky wage setting in Erceg et al. (2000), this simplified wage-setting equation assumes a perfectly competitive labor market, and thus there is no wage markup if the firms are allowed to reset the wage contract. In addition, the trapped wage is on the steady-state level, which can be taken as the optimal wage setting before a shock has happened.

Another property of this equation is that the implied wage markup is not constant. To see this, we can simply calculate

$$\frac{w_t}{MRS_t} = \left(\frac{L_t^{\psi}}{C_t^{-\sigma}}\right)^{\theta_w - 1} w^{1-\theta_w} = MRS_t^{\theta_w - 1} w^{1-\theta_w}.$$

The wage markup increases a lot when the MRS_t approaches 0, which means the wage markup can absorb much of the decrease of the MRS_t in this regime and keeps the change of the real wage smooth. This mechanism is shown by Justiniano et al. (2010) to be important in accounting for the role of the investment shock in the business cycle.

2.5 Government

The government performs the monetary and fiscal policy. The central bank follows a standard Taylor rule,

$$R_t = \left(\pi^* r \left(\frac{\pi_t}{\pi^*}\right)^{\psi_\pi} \left(\frac{Y_t}{Y_t^n}\right)^{\psi_y}\right)^{1-\rho_R} R_{t-1}^{\rho_R} \exp(\varepsilon_{R,t}) \tag{20}$$

π^* is the targeted inflation rate of the central bank, and r is the steady-state real interest rate. The nominal interest rate is set according to the deviation of inflation π_t from the targeted inflation rate and of output Y_t from the natural output Y_t^n. Finally, $R_{t-1}^{\rho_R}$ smooths the change of the nominal interest rate, and $\varepsilon_{R,t}$ is the monetary policy shock, with $\varepsilon_{R,t} \sim$ i.i.d. $\mathcal{N}(0, \sigma_R^2)$ across time.

The fiscal policy part includes a government budget constraint and an exogenous government spending process.

$$P_t G_t + R_{t-1} B_{t-1} = B_t + T_t \tag{21}$$

Government spending, G_t, follows an output stabilization rule and an exogenous government spending process (Leeper et al., 2010).

$$G_t = \left(\frac{Y_t}{Y}\right)^{-\phi_g} \mu_{g,t} \tag{22}$$

$\mu_{g,t}$ follows,

$$\ln \mu_{g,t} = (1 - \rho_g) \ln \mu_g + \rho_g \ln \mu_{g,t-1} + \varepsilon_{g,t}, \ \ \varepsilon_{g,t} \sim \text{i.i.d. } \mathcal{N}(0, \sigma_g^2). \tag{23}$$

2.6 Resource Constraint

First we define the aggregate variables from the integration of households' optimal choices,

$$\begin{aligned} &C_t = \int_0^1 c_t(j)\, dj, \ I_t = \int_0^1 i_t(j)\, dj, \ L_t = \int_0^1 \ell_t(j)\, dj, \\ &K_t = \int_0^1 k_t(j)\, dj, \ B_t = \int_0^1 b_t(j)\, dj, \end{aligned} \tag{24}$$

Since capital utilization u_t is only associated with the real interest rate of capital in equation (10), it is the same across all households. After integrating on the variables of $k_t^s(j)$ and $n_t(j)$ on the firms' side, and then applying the equilibrium condition, we can get

$$\begin{aligned} Y_t &= C_t + I_t + G_t + a(u_t)K_{t-1} \\ K_t^s &= u_t K_{t-1} \\ N_t &= L_t \end{aligned} \tag{25}$$

The government bond market clears according to Walras' law.

3 Equilibrium Solution Method

This section introduces the investment demand shock to the model and the solution method. The solution requires multiple parallel models—one for solving the endogenous investment demand, and two for solving the natural output.

3.1 Investment Demand Shock and Second-stage Model

The starting point is the investment level decided by the baseline model, which is denoted as I_t^*. After the equilibrium is established in the baseline model, the investment demand shock $\varepsilon_{d,t}$ is revealed, causing total investment to increase exogenously. Specifically, $\mu_{d,t}$ is the exogenous investment demand process, following

$$\ln \mu_{d,t} = (1-\rho_d)\ln \mu_d + \rho_d \ln \mu_{d,t-1} + \varepsilon_{d,t}, \;\; \varepsilon_{d,t} \sim \mathcal{N}(0, \sigma_{d,t}^2) \tag{26}$$

and the total investment becomes

$$I_t = \mu_{d,t} I_t^*. \tag{27}$$

As all households are representative, the increase in aggregate investment is equivalent to each household increasing its investment i_t^* by $\mu_{d,t}$. Therefore, we can specify the new optimal problem for each household.

$$\begin{aligned} &\max_{c,b,k,l,u} \; \mathbb{E}_t \sum_{k=0}^{\infty} \beta^k \left(\frac{(c_{t+k}/\gamma^{t+k})^{1-\sigma} - 1}{1-\sigma} - \chi \frac{\ell_{t+k}^{\psi+1}(j)}{\psi+1} \right) \\ &\text{s.t. } c_t + \mu_{d,t} i_t^* + \frac{b_{t+1}}{R_{t+1}P_t} = r_t^k k_{t-1} u_t - a(u_t) k_{t-1} + \frac{b_t}{P_t} + w_t \ell_t - \frac{\tau_t}{P_t} + \frac{d_t}{P_t} \\ &k_{t+1} = (1-\delta)k_t + \mu_{s,t} \left[1 - S\left(\frac{\mu_{d,t} i_t^*}{i_{t-1}} \right) \right] \mu_{d,t} i_t^* \end{aligned} \tag{28}$$

The difference between this optimal problem and the one in Section 2 lies in the control variables. Equation (28) states that investment is no longer a control variable.

Instead, it is decided by the investment law described in equation (27), and thus we do not have the first-order condition with respect to investment. This kind of specification is not common, because it actually says households are making the investment decision irrationally—the investment is not decided by the first-order condition. However, Nieri and Ragot (2025) provides the micro-foundation of such shock, which is generated from firms' idiosyncratic productivity shocks and the complementary movement between firms.

Following the exogenous investment required by the household, the economy should solve the equilibrium again, which will be finally used to calculate the likelihood. The reason why the investment demand shock cannot be compounded with the baseline model is that the household will adjust their investment according to the realization of the shock otherwise.

3.2 Frictionless Parallel Models

To solve the natural output used in the Taylor's rule, two frictionless parallel models should be introduced, one of which corresponds to the baseline model, and the other of which corresponds to the second-stage model. The two frictionless parallel models are nearly the same, except for the setting related to the investment demand shock.

Inside each frictionless parallel model, all nominal variables are excluded, and the wage sticky parameter θ_w is set to zero so that the real wage equals the marginal rate of substitution between hours worked and consumption.

The relations between all models are illustrated in Figure 1. Basically, starting from the frictionless baseline model, we solve the endogenous investment and natural output. The endogenous investment is then sent to the frictionless second-stage model, and the natural output to the sticky baseline model. Next, another endogenous investment is solved from the sticky baseline model and another natural output from the frictionless second-stage model. Finally, combining the new endogenous investment and natural output, I solve the second-stage model.

3.3 Detrending and Log-linearization

As the model is endowed with a growth trend, the detrended state variables should be defined before solving the steady state. Specifically, define

$$\tilde{Y}_t = \frac{Y_t}{\gamma^t},\ \tilde{C}_t = \frac{C_t}{\gamma^t},\ \tilde{I}_t = \frac{I_t}{\gamma^t},\ \tilde{K}_t = \frac{K_t}{\gamma^t},\ \tilde{K}_t^s = \frac{K_t^s}{\gamma^t},\ \tilde{w}_t = \frac{w_t}{\gamma^t} \tag{29}$$

After rewriting the first-order conditions in terms of the detrended model, we can solve the steady state and the log-linearized model. The detailed derivation is put in Appendix A.

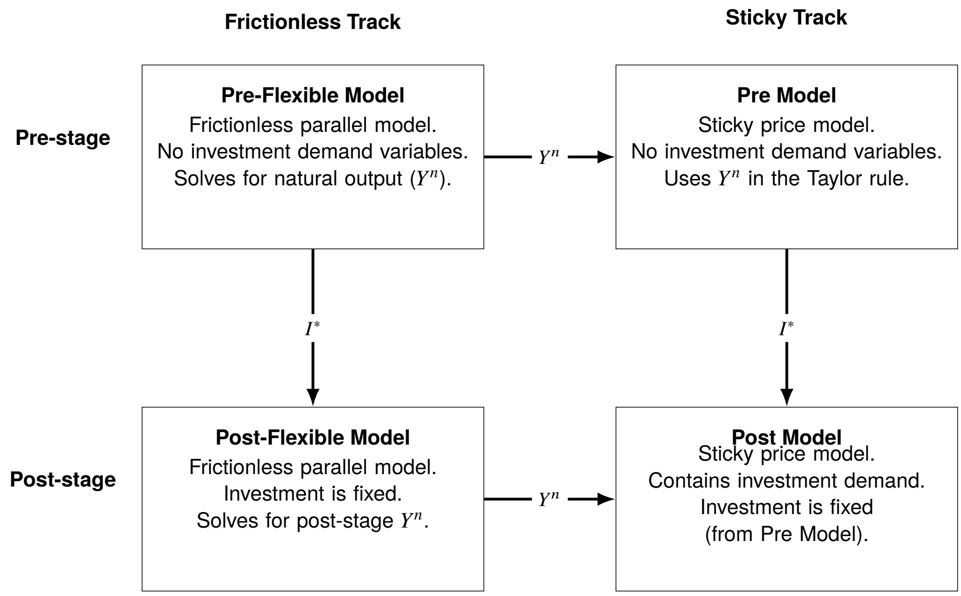

Figure 1: Link between the baseline model, the second-stage model with investment demand shock, and the frictionless parallel models used to compute endogenous investment and natural output.

4 Estimation

This section describes the estimation method, including data selection, priors and posteriors of parameters. The methodology follows Smets and Wouters (2007) and Justiniano et al. (2010).

4.1 Data

The data set contains U.S. data on GDP growth rate, inflation rate, the federal funds rate, hours worked, investment growth rate, and government spending growth rate. The data period is from 1983:Q1 to 2002:Q4. The period selection is based on three considerations. First, since the economic model assumes a constant linear growth trend, we may face the problem of changing growth rates if we use a data set over long periods. Second, the data is selected to be comparable with Justiniano et al. (2010), who used the data from 1954:Q3 to 2004:Q4. Third, the data period avoids the periods where the economy faced the zero-lower bound constraint, as the model lacks a clear picture of monetary policy in such regimes.

All data are collected from the FRED database, and are processed to represent per-capita levels because there is no detailed modeling of population growth in the baseline model. The detail of the data process is described in Appendix B.

$$\begin{bmatrix} \Delta \ln GDP_t \\ \Delta \ln P_t \\ FEDFUNDS_t \\ \ln HOURS_t \\ \Delta \ln INV_t \\ \Delta \ln GOV_t \end{bmatrix} = \begin{bmatrix} 100 \ln \gamma \\ 400\pi_{ss} \\ 400R_{ss} \\ \bar{\ell} \\ 400 \ln \gamma \\ 400 \ln \gamma \end{bmatrix} + \begin{bmatrix} \hat{y}_t - \hat{y}_{t-1} \\ \hat{\pi}_t \\ \hat{R}_t \\ \hat{\ell}_t \\ \hat{i}_t - \hat{i}_{t-1} \\ \hat{g}_t - \hat{g}_{t-1} \end{bmatrix} + \Sigma_t^{meas} \tag{30}$$

Equation (30) describes the relationship between the observed data and the state variables, which is called the observation equation. π_{ss} and R_{ss} are steady-state values of these variables, and variables with hats indicate the log-linearized variables. π_{ss} and R_{ss} are multiplied by 4 to indicate the year-to-year inflation rate and the federal funds rate. Moreover, because I use the demeaned $HOURS_t$, $\bar{\ell} = \ell_{ss} - \mu_{hours}$, where ℓ_{ss} is the steady-state labor input level and μ_{hours} is the mean of the data.

Σ_t^{meas} is a 6×1 vector, containing the measurement error for each data variable. The measurement error helps the model to adjust the number of shocks flexibly, while the stochastic singularity problem may be raised with fewer shocks than the data (Ruge-Murcia, 2007). The value of the measurement error is set as 4% of the variance of each series, meaning that there is a small part of variance in the data caused by the data collection process.

Compared with Smets and Wouters (2007) and Justiniano et al. (2010), I add the data of government spending, because there are fewer shocks in my model than theirs and the government spending shock will dominate the fluctuation otherwise. In addition, the data on consumption and wages are dropped, as more shocks and mechanisms are required for the model to fit these additional data. Nevertheless, the result of the expanded model will be examined in section 6.

4.2 Prior and Posterior Distributions

To facilitate the setting of the priors, I reparametrize some of the parameters. Specifically,

$$\begin{aligned} \bar{\gamma} &= 100(\gamma - 1) \approx 100 \ln \gamma, \\ \bar{\pi} &= 400\pi_{ss}, \\ \bar{r} &= 400 \left(\frac{\gamma}{\beta} - 1 \right). \end{aligned} \tag{31}$$

The steady state nominal interest rate, R_{ss}, is calculated as $\bar{\pi} + \bar{r}$ according to equation (31). Additionally, there are some parameters that suffer from weak identification problems, and therefore I set their values following Smets and Wouters (2007); Leeper et al. (2010); Nieri and Ragot (2025). The elasticity of substitution between intermediate goods ν_p is set as 6, indicating that the steady-state markup is 1.2. The depreciation rate of capital δ is set as 0.025, and the steady ratio of government spending to output s_g is set as 0.2.

The details about the priors and posteriors are summarized in table 1, where the posterior is based on the Metropolis-Hastings algorithm with a sample size of 100,000 and a burn-in rate of 0.5. The priors of the parameters are relatively flat so that fewer restrictions are imposed on the posteriors.

Table 1: Prior distributions and posterior summaries (mean, std, 5%, 95%) for parameters in the baseline estimation.

	Prior			Posterior			
Parameter	Dist.	Mean	Std	Mean	Std	5%	95%
σ	Gamma	2.00	0.50	2.56	0.37	1.99	3.19
ψ	Gamma	2.00	0.50	1.77	0.26	1.37	2.23
κ_p	Beta	0.30	0.10	0.43	0.07	0.32	0.55
α	Normal	0.30	0.05	0.41	0.02	0.38	0.44
θ_w	Beta	0.50	0.20	0.50	0.10	0.35	0.67
$\bar{\ell}$	Normal	0.00	2.00	0.96	1.69	-1.82	3.77
$\bar{\gamma}$	Normal	0.40	0.20	0.33	0.03	0.27	0.37
$\bar{r}$	Gamma	0.50	0.50	1.80	0.44	1.07	2.49
$\bar{\pi}$	Gamma	7.00	2.00	4.05	0.69	3.05	5.36
ψ_π	Normal	1.50	0.25	1.87	0.18	1.60	2.18
φ_i	Gamma	4.00	1.50	8.35	1.98	5.40	11.79
ψ_u	Beta	0.50	0.15	0.09	0.03	0.04	0.14
ϕ_g	Gamma	0.07	0.05	0.04	0.01	0.02	0.06
ψ_y	Gamma	0.50	0.25	0.28	0.13	0.10	0.52
ρ_a	Beta	0.60	0.20	0.94	0.02	0.89	0.98
ρ_g	Beta	0.60	0.20	0.99	0.00	0.99	0.99
ρ_R	Beta	0.60	0.20	0.76	0.04	0.70	0.82
ρ_s	Beta	0.60	0.20	0.99	0.01	0.97	1.00
ρ_d	Beta	0.60	0.20	0.87	0.04	0.80	0.93
σ_a	InvGamma	0.50	4.00	0.45	0.06	0.36	0.56
σ_g	InvGamma	0.50	4.00	0.89	0.07	0.78	1.01
σ_R	InvGamma	0.50	4.00	0.24	0.03	0.20	0.29
σ_s	InvGamma	0.50	4.00	1.41	0.35	0.89	2.04
σ_d	InvGamma	0.50	4.00	1.35	0.16	1.10	1.62

φ_i is the parameter in investment adjustment function, which is defined in equation (64) in Appendix A. ψ_u is the parameter in the capital utilization adjustment cost function defined in equation (67). Moreover, κ_p is the coefficient in the Phillips curve, defined in equation (71).

There are a few posteriors worth examining. The first one is ψ_u, whose posteriors differ quite a lot across the literature. Smets and Wouters (2007) estimates ψ_u at about 0.5, Justiniano et al. (2010) gives about 0.8, and Section 6 gives about 0.1, even though the model structure is quite similar. The second one is ρ_s, as it is very close to the unit root. This extreme value may be a result of lacking enough shocks to interpret the volatility in the data, as the result Section 6 shows a more temperate persistence in IST shock. Finally, κ_p and θ_w are around 0.5, which indicates the Calvo parameter is near 0.5 for the final goods, and 0.2 for the wage. This means 50% of final goods producers can reset their price, and 80% of households can reset their wage. I flag these problematic posteriors here and will investigate their implications for other results in the next section.

5 Results

This section introduces the estimation results from the baseline model, in which the central part is the forecast error variance decomposition (FEVD). I begin with a moments comparison between the estimated model and the data.

5.1 Moments

Table 2 shows the comparison of mean and variance implied by the model and the data. Though most of the moments are close, the variance of the hours worked is extremely high compared with the data. To investigate this result, we can look into the log-linearized equations to write the labor input by state variables out of the labor market.

$$\hat{N}_t = \frac{\hat{mc}_t + \alpha\hat{k}_t^s + \hat{\mu}_{a,t} - \theta_w\sigma\hat{c}_t}{\alpha + \theta_w\psi} \tag{32}$$

I decompose the stationary variance of $\hat{N}_t$ into the four parts laying on the numerator. Specifically, the stationary variance is computed by

$$\mathrm{Var}(\bar{s}) = \mathrm{Var}(\Phi\bar{s} + R\varepsilon) = \Phi\mathrm{Var}(\bar{s})\Phi' + R\mathrm{Var}(\varepsilon)R',$$

and then R^2 is computed according to equation (32) and the variance of each state variables. The result is summarized in Table 3.

In such decomposition, the high volatility of hours worked mainly comes from the volatile consumption log-deviation (level) $\hat{c}_t$. Though in the data, the variance of $\hat{c}_t$ is just about 2.8, the model implied counterpart is 39.23 as shown in Table 4. While it is normal to think that consumption is volatile in the baseline estimation, because the consumption data is not used in the data set, Table 4 also shows that this problem persists

Table 2: Model-implied and data moments (mean and variance) for the baseline observables (output, inflation, interest rate, hours, investment, and government spending).

	Mean		Variance	
Observable	Model	Data	Model	Data
Output	0.33	0.56	0.67	0.34
Inflation	4.05	3.08	5.09	2.16
Interest rate	7.15	6.05	8.33	5.01
Hours	0.96	0.00	31.86	6.37
Investment	0.33	0.57	3.87	2.53
Gov. spending	0.33	0.31	0.82	0.85

Table 3: Decomposing model-implied hours volatility. $R^2 = \text{corr}(N_t, x_t)^2$ from regressing model-implied hours on each component x_t, i.e. the share of $\text{Var}(N_t)$ linearly accounted for by co-movement with that component. Moments are theoretical (unconditional) second moments, $\Sigma = \Phi\Sigma\Phi' + R\,Se\,R'$.

Component	$\text{corr}(N_t, x_t)$	R^2 (%)
Consumption (wealth effect), $-\omega\sigma c_t$	+0.907	82.29
Marginal cost, mc_t	+0.061	0.37
Capital services, $\alpha(k_t + u_t)$	-0.030	0.09
Technology, a_t	+0.087	0.76

in the workhorse model in Smets and Wouters (2007). I compute this statistic from the replication package of Smets and Wouters (2007) directly, the value reaches to 33.75 as well.

Table 4: Model-implied vs. data variance of the *level* of consumption and hours worked. Data variance of consumption level is computed by the cyclical term detrended by OLS constant and time-liear term. The Smets and Wouters (2007) column reports the theoretical unconditional moments of the original Smets–Wouters (2007) model. If we assume state variables follow AR(1) processes, the gap between the model levels and the data is driven by persistence: $\mathrm{Var}(\mathrm{level}) = \mathrm{Var}(\mathrm{growth})/[2(1-\rho_1)]$.

Variable	Smets and Wouters (2007)	Baseline model	Data
Consumption (c)	33.75	39.23	2.80
ρ_1	0.993	0.996	0.901
Hours worked (N)	8.85	32.22	6.29
ρ_1	0.975	0.993	0.907

However, the variance of labor input differs quite a lot. This is because Smets and Wouters (2007) adds a wage markup shock $\mu_{w,t}$ between the marginal rate of substitution and the real wage

$$\hat{\pi}_t^w = \beta \mathbb{E}_t \hat{\pi}_{t+1}^w - \kappa_w \underbrace{\left(\hat{w}_t - \hat{mrs}_t + \frac{1}{\kappa_w} \mu_{w,t} \right)}_{\text{wage markup}}.$$

By adding such shock, wage is less tied with MRS_t, and thus it has more freedom to fit with the data. In 6, the estimation based on the Smets and Wouters (2007) model shows a much better fit for the variance of hours worked, and the FEVD analysis in Table 7 also shows the strong influence of the wage markup shock on hours worked and wage growth.

Nevertheless, from the wage Phillips curve equation, it is natural to think that κ_w will play an important role on the connection between consumption and labor input. Actually, by computing κ_w under posterior in Section 6, it is only tiny 0.006. But if we drop the wage markup shock, the weak link between MRS_t and wage effectively let the wage to be nearly static, and it is why even if the parameter θ_w exists in the simplified wage setting equation, its posterior still stays about 0.5.

5.2 Forecast Error Variance Decomposition

Table 5 is the unconditional FEVD computed by a 400-iteration algorithm because directly solving the Lyapunov equation is not numerically stable in this model. As illustrated in the table, both IST shock and investment demand shock have important effects on the business cycle. The sum of these two shocks can account for about 65% of the forecast error of the output, 45% of inflation, and 90% of hours worked. This result is similar to Justiniano et al. (2010), who claimed that the IST shock contributes about 50% of the forecast error of the output and 60% of hours worked.

Separately, investment demand shock alone contributes about 40% of the forecast error of the output, and about 50% of the investment. However, investment demand shock contributes much less for the hours worked than IST shock.

Table 5: Unconditional FEVD (%) for the baseline estimation, by government spending (ε_g), IST (ε_s), monetary policy (ε_R), technology (ε_a), and investment demand (ε_d) shocks.

Observable	ε_g	ε_s	ε_R	ε_a	ε_d
Output	4.63	15.55	11.00	27.12	41.69
Inflation	0.32	39.26	12.87	40.14	7.41
Interest rate	0.37	55.06	5.48	31.22	7.87
Hours	4.34	90.19	0.08	3.56	1.83
Investment	0.11	36.23	0.15	12.04	51.48
Gov. spending	99.87	0.02	0.01	0.04	0.06

The first reason is the high autocorrelation of IST shock. As shown in Table 1, ρ_s is quite close to the unit root, while $\rho_d = 0.87$ in the posterior. Second, IST shock triggers a hump-shape adjustment of investment due to the investment adjustment cost. Therefore, its peak arrives late and thus more persistent. By contrast, investment demand shock doesn't have such effect, because it changes the investment exogenously. Finally, the substitution effect of IST shock is strong. As shown in Figure 2, IST shock causes the instant negative wage response but the positive labor input response. The story behind it is that the shock moves the labor supply curve toward the right, reflecting the incentive to save more today because of the cheap investment. The negative response of consumption and the nearly static output confirm the intertemporal substitution effect, as they purely increase saving and decrease consumption. On the other side, the investment demand shock shows less substitution effect. As the wage and labor input increase simultaneously, it is the movement of the labor demand curve that dominates the labor market. Though consumption decreases as well, output increases a lot, which shows that households haven't shifted consumption to investment completely. This is the wealth effect discussed in Nieri and Ragot (2025) and Greenwood et al. (1988): higher investment means higher output in the future, and thus higher lifetime wealth.

Nevertheless, Justiniano et al. (2010) uses the frequency based FEVD. To faciliate comparing, Table 6 shows the FEVD result based on the same methodology.

The result shows that the aggregate technology has a much stronger effect on the

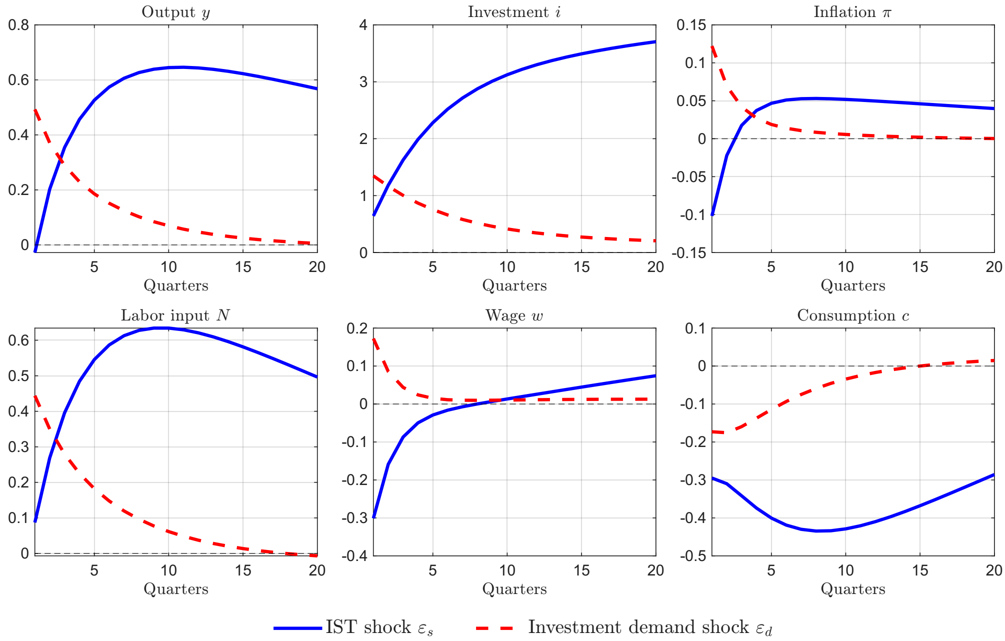

Figure 2: Impulse responses of output, investment, inflation, labor input, wage, and consumption to one-standard-deviation IST and investment demand shocks (baseline estimation; horizon: 20 quarters).

Table 6: Frequency-domain FEVD (%) for the business-cycle band (6–32 quarters), with measurement error included (baseline estimation; 500 frequency bins).

Observable	ε_g	ε_s	ε_R	ε_a	ε_d	meas
Output	1.44	28.15	2.16	57.59	9.77	0.89
Inflation	0.13	16.26	17.73	59.52	5.06	1.29
Interest rate	0.16	22.76	9.76	58.55	6.33	2.45
Hours	2.68	49.48	1.49	23.25	16.69	6.42
Investment	0.19	54.18	0.18	28.69	15.72	1.03
Gov. spending	95.93	0.02	0.00	0.04	0.01	4.00

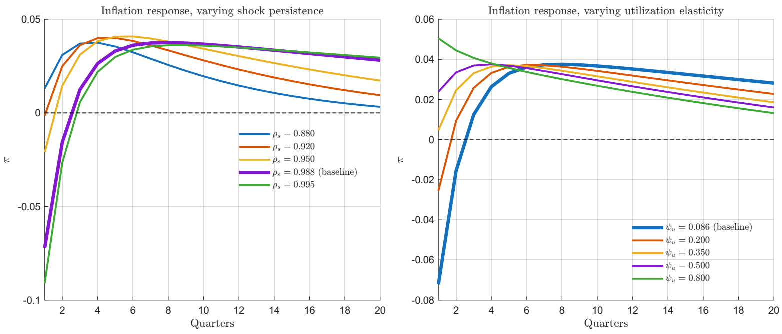

Figure 3: The sign of the inflation response to an IST shock depends on the estimated shock persistence ρ_s and the capital-utilization elasticity ψ_u.

business cycle, which accounts for about 50% for output, inflation and nominal interest rate. IST shock contributes for 28.15% of output, 16.26% of inflation, 54.14% of investment, and 49.48% of hours worked. Meanwhile, investment demand shock is less important, accounting for 9.77% of output, 5.06% of inflation, 15.72% of investment and 16.69% of hours worked.

Compared with the unconditional FEVD analysis in Table 2, the frequency FEVD restricts the impact to between 6 and 32 quarters. From the IRF plot in Figure 2, we can observe that the effect caused by investment demand shock almost converges to zero after 6 periods, which explains why it contributes less in frequency FEVD result.

5.3 Inflation Impact

Another interesting point in Figure 2 is that the IST shock results in deflation in the first periods, while the investment demand shock generates inflation. Although this difference is documented in Nieri and Ragot (2025), which can be explained by the competition between substitution effect and wealth effect introduced before, Justiniano et al. (2010) and Section 6 in this paper propose the contrary result—IST shock raises deflationary effect. Comparing the posterior, ρ_s and ψ_u show a big difference. Figure 3 tests the inflation response to IST shock under various values of ρ_s and ψ_u, where ψ_u is defined from the household first-order condition of capital utilization rate (10),

$$\hat{r}_t^k = \frac{a''(1)}{a'(1)}\,\hat{u}_t \equiv \frac{\psi_u}{1-\psi_u}\,\hat{u}_t.$$

Under high ρ_s and low ψ_u, the deflationary result is produced. However, inflationary result can be shown if ρ_s is below 0.92 or ψ_u is above 0.32, keeping all other parameters unchanged. To understand the result, we can write the inflation as the summation of

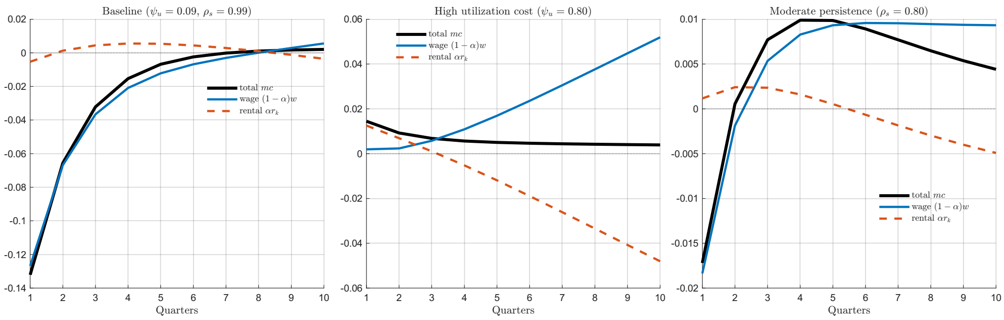

Figure 4: Decomposing the marginal-cost response to an IST shock: a wage channel against a capital-rental channel.

present discounted value of marginal cost,

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa_p \widehat{mc}_t \implies \hat{\pi}_t = \kappa_p \underbrace{\sum_{j=0}^{\infty} \beta^j \mathbb{E}_t [\widehat{mc}_{t+j}]}_{=\text{PDV}(mc)} . \tag{33}$$

Furthermore, there are two channel to influence the marginal cost—wage and capital rental rate, shown in Figure 4. The first shows that under low ψ_u, marginal cost is mainly accounted by the wage channel, while capital rental rate has only small effect in the first few periods. To understand this, we associate capital rental rate with firm's cost minimization decision (70) and households' capital utilization decision (67),

$$\begin{cases} \hat{r}_t^k = \hat{w}_t + \hat{N}_t - \hat{K}_{t-1} - \hat{u}_t \\ \hat{r}_t^k = \frac{\psi_u}{1-\psi_u} \hat{u}_t \end{cases} \implies \hat{r}_t^k = \psi_u \left(\hat{w}_t + \hat{N}_t - \hat{k}_{t-1} \right) . \tag{34}$$

Equation (34) shows $\hat{r}_t^k$ will only move ψ_u fraction of $\hat{w}_t + \hat{N}_t$, as $\hat{k}_{t-1}$ is predetermined. First, under baseline result, $\hat{w}_t$ and $\hat{N}_t$ move to the opposite direction, which cancels out the absolute value of the difference. Second, the posterior of ψ_u is only about 0.1, dampening the magnitude a lot. However, the second panel of Figure 4 shows the capital rental rate channel becomes important as ψ_u becomes large, leading to the inflationary result. To understand this change, we can recall the definition of ψ_u again. As $\frac{a''(1)}{a'(1)} = \frac{\psi_u}{1-\psi_u}$, increase of ψ_u implies $a''(1)$ becomes relatively larger. This means the adjustment cost function is steeper around the steady state. When the IST shock creates the increasing labor supply, marginal productivity of capital increases and thus capital demand increases. However, capital supply can hardly adjust due to the expensive adjustment cost. Therefore, the strong movement of the capital demand curve results in the high rental rate in the first few periods.

Understanding the role of ψ_u helps the further analysis about ρ_s. Since ψ_u is small in baseline posterior, we can focus on the wage channel. When ρ_s decreases, the IST

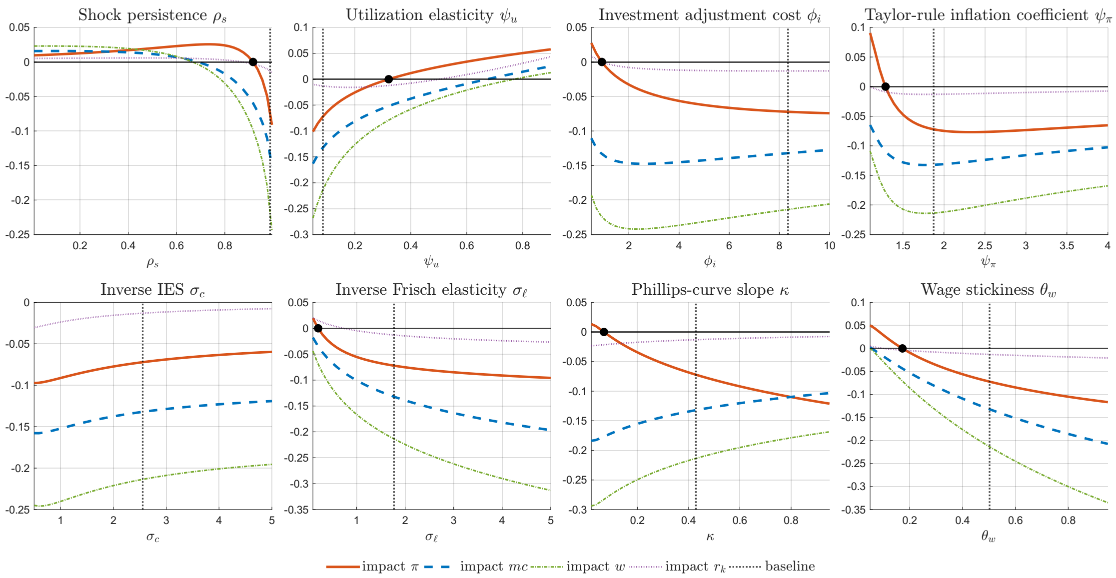

Figure 5: Sensitivity of the sign of the impact inflation response to an IST shock. Each panel varies one parameter over a wide grid

shock will persists shorter period once it arrives. Suppose IST shock is positive, then household will save more for shorter period when the investment is cheap, and thus wage will decrease for shorter time. As the current inflation is the summation of future marginal cost shown in equation (33), it will be positive once the summation of negative initial terms is outweighed by the positive tail, as shown in the third panel of Figure 4.

It is worth stressing that the inflation impact under the IST shock is also sensitive to many other parameter values, such as the wage sticky parameter and the Phillips curve parameter, shown in Figure 5. Although the sign of the inflation impact is not robust to the parameter value choice, the sign of the wage impact shows a stable result in this exercise. However, I acknowledge the limitation of such an exercise: it alters parameters one by one, but cannot investigate the effect of simultaneous movement among several parameters. In fact, Justiniano et al. (2010) shows the positive wage impact under IST shock.

On the other side, Figure 6 shows the evidence about the robust inflationary effect of investment demand shock. The left panel shows both wage and capital rental rate increase under investment shock. This is because the shift of labor demand curve pushes up both labor input and wage, and thus capital rental rate increases by equation (34). The right panel shows that the inflation impact is positive under the investment demand shock across all 500 selected draws. Moreover, as shown in Figure 7, the parameter sweep exercise confirms the robustness of this effect, and the wage impact remains positive during every single parameter sweep.

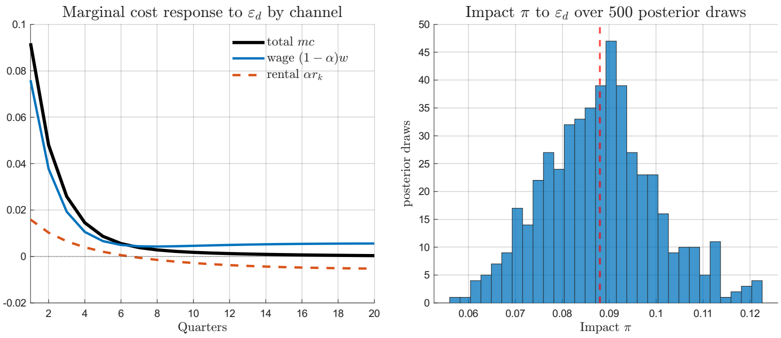

Figure 6: Robustness of the inflationary response to an investment-demand shock. *Left panel.* Channel decomposition of the real marginal-cost response to a unit investment-demand shock at the baseline posterior mean. *Right panel.* Impact inflation $\hat{\pi}_1$ in response to a unit investment-demand shock across 500 posterior MH draws that are selected linarly across index among MH samples after burn-in.

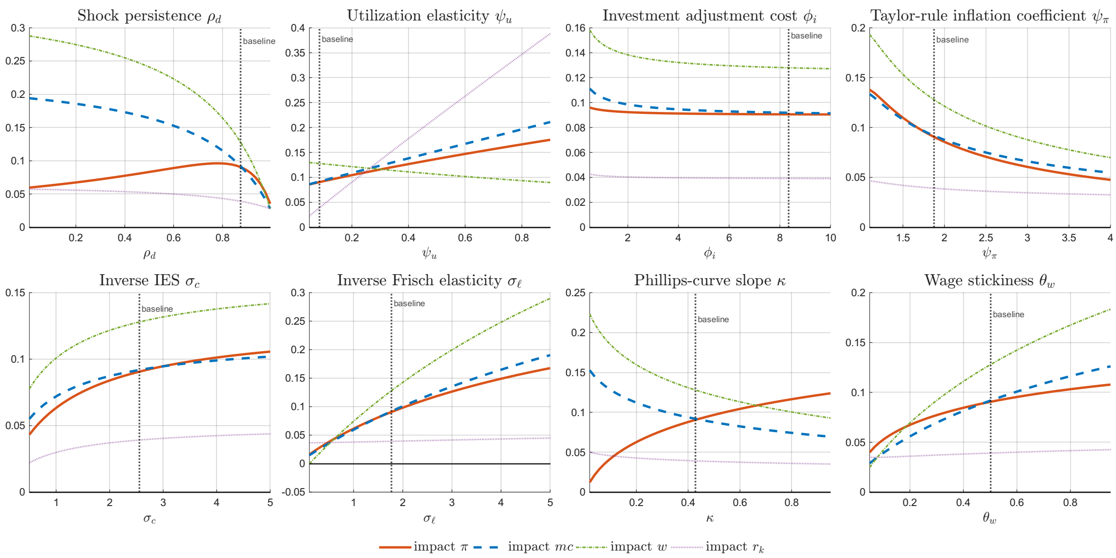

Figure 7: Robustness of the sign of the inflation response to an investment-demand shock. Each panel varies one parameter over a wide grid—the shock persistence ρ_d, the capital-utilization elasticity ψ_u, the investment adjustment cost φ_i, and the Taylor-rule inflation coefficient ψ_π—holding all other parameters at the baseline posterior mean

5.4 Consumption Impact

One of the attractions of the investment demand shock is its ability to generate comovements between consumption, investment and inflation, while the IST shock is usually unable to reproduce such a result. Nieri and Ragot (2025) shows that the positive comovement shows up under certain calibrations, and one of the quests of this paper is to test whether the estimation supports such a result.

However, as shown in the IRF plot, Figure 2, the consumption impact is negative. Moreover, altering a single parameter value cannot reproduce a positive consumption impact either—such a result relies on certain parameter combinations. To figure out which parameter combination is needed, I align some model settings to Nieri and Ragot (2025), and simultaneously change three parameters that are important in theoretical proof—Phillips-curve slope κ_p, the Taylor-rule inflation coefficient ψ_π and the wage-rigidity parameter θ_w. Result in Figure 8 shows that, generally, if all three parameters are low enough, then consumption impact can be positive [2].

The result can be understand as follows. In short run, investment demand shock pushes up aggregate demand, but in long run accumulating more capital. Instead of investment, imagine that the capital increases directly. This simply makes households more wealthy, and thus they will consume more today. So, how to let the investment demand shock be analogous to direct capital increase? We can shut down the short-run effect by letting the economy to be more sticky—lower θ_w suppressing wage, lower κ_p suppressing inflation, and lower ψ_π suppressing nominal interest rate. They can then form a chain to suppress intertemporal substitution effect. If the wage increases only mildly, marginal cost will not increase much; if the Phillips curve is flat, inflation will not increase much; and if the Taylor rule is inactive to inflation, then the nominal interest rate will be stable; finally, since the real interest rate is stable, households will not shift today's consumption to tomorrow.

5.5 Historical Decomposition

To investigate the shock effects along the historical data, I show the historical decomposition in Figure 9, where the fluctuation of the fundamental macroeconomic data are decomposed into the fluctuation of structural shocks.

Compared with the IST shock, the investment demand shock is shown to be more volatile in the HD of output, which often switches direction from quarter to quarter. Also, it behaves more procyclically, nearly moving in the same direction as output and inflation.

[2] Some may question about the indeterminate area in Figure 8, since the textbook new Keyasian model can be uniquely solven once $\psi_\pi > 1$. However, the problem becomes more complicated when capital and investment are added into the model, and Sveen and Weinke (2005) provides a detailed theoretical analysis about it.

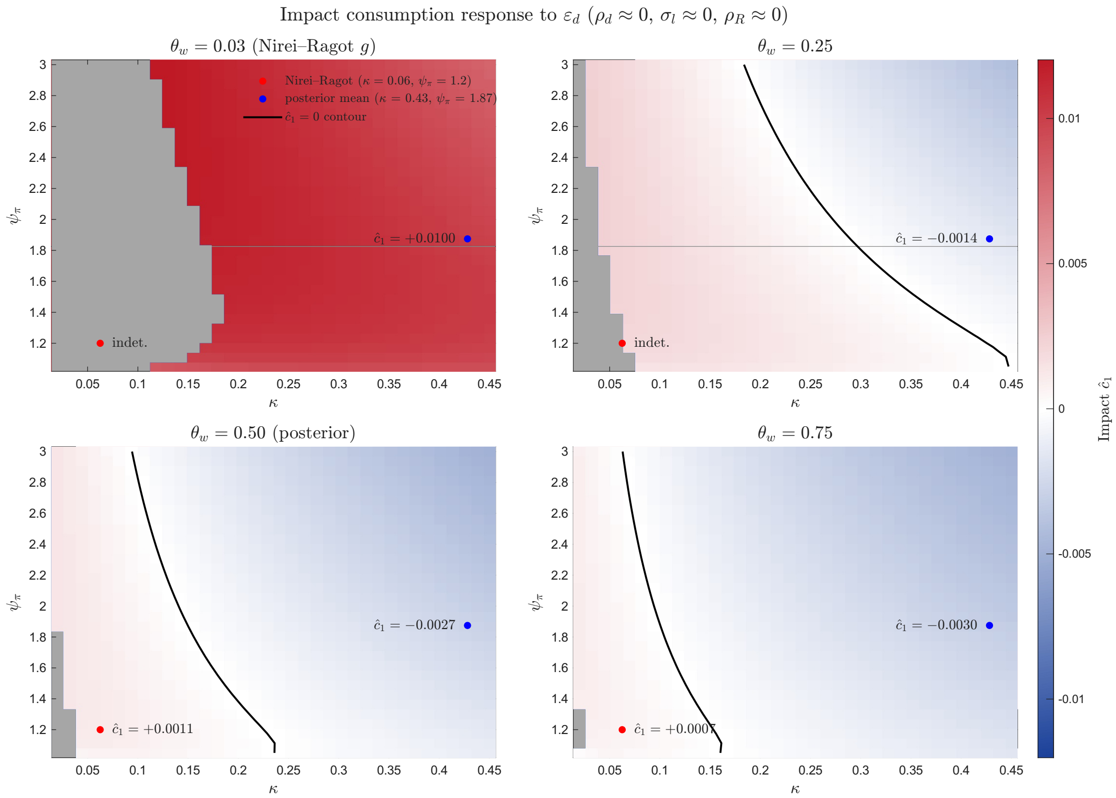

Figure 8: Impact consumption response $\hat{c}_1$ to a unit investment-demand shock as a function of the Phillips-curve slope κ_p and the Taylor-rule inflation coefficient ψ_π four values of the wage-rigidity parameter θ_w. The remaining parameters are held at the calibration in Nieri and Ragot (2025): an i.i.d. shock ($\rho_d \approx 0$), a non-inertial pure-inflation rule ($\rho_R \approx 0$), and near-linear labor disutility ($\sigma_l \approx 0$). Gray cells are indeterminate configurations.

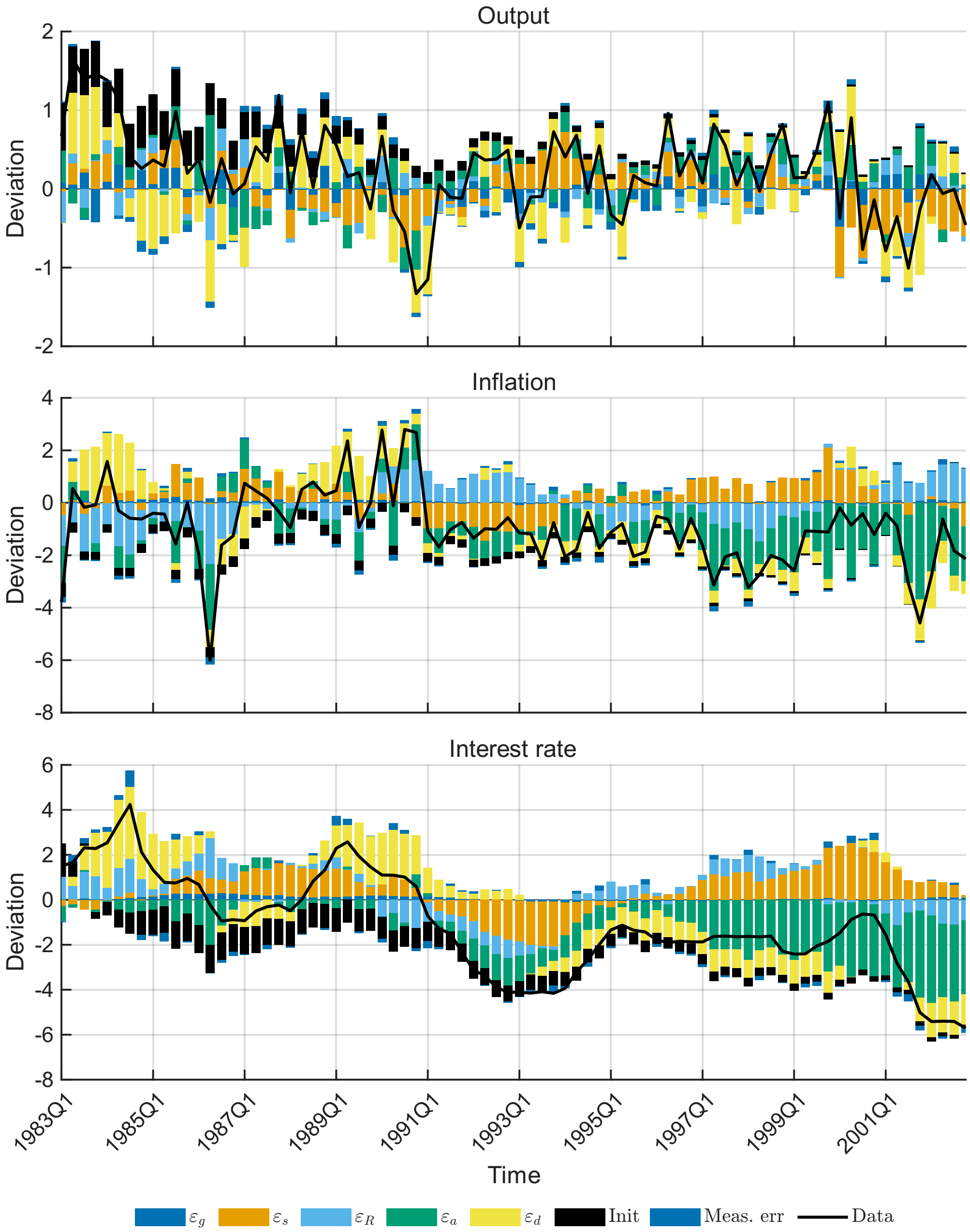

Figure 9: Historical decomposition (Kalman smoother) of demeaned output growth, inflation, and nominal interest rate into contributions from structural shocks (including initial conditions and measurement error) in the estimated baseline model.

During the oil crisis in 1990-1991, both the IST shock and the investment demand shock show a very negative influence on output. After 1995, the aggregate productivity shock becomes the main driving force of inflation and the nominal interest rate.

6 Robustness Test

Three alternative settings are used to test the baseline result regarding the investment demand shock—the first one uses the standard Smets and Wouters (2007) model; the second discusses the case where the investment demand shock is i.i.d.; the last one further differentiates the IST shock and the marginal efficiency of investment (MEI) shock, following Justiniano et al. (2011).

6.1 Smets–Wouters-style Model

In this section, the baseline model is expanded following Smets and Wouters (2007) to address the underlying concerns of oversimplification. Specifically, I add a habit formation term in the utility function and a monopolistic competition structure in the labor market. In addition, a preference shock, a price markup shock and a wage markup shock are added as well. In terms of data, the consumption and wage growth rates are added.

The utility function of households becomes

$$\mathbb{E}_t \left[\sum_{s=0}^{\infty} \beta^s \mu_{b,t+s} \left(\ln(C_{t+s} - hC_{t+s-1}) - \chi \frac{L_{t+s}(j)^{1+\psi}}{1+\psi} \right) \right], \tag{35}$$

where $\mu_{b,t}$ is the intertemporal preference shock, following

$$\ln \mu_{b,t} = (1-\rho_b)\mu_b + \rho_b \ln \mu_{b,t-1} + \varepsilon_{b,t}, \ \ \varepsilon_{b,t} \sim \text{i.i.d. } N(0, \sigma_b^2). \tag{36}$$

It is worth to mention that the consumption part in the utility function becomes logarithmic instead of the more general CRRA form. This is because with habit formation and separable utility function between C_t and L_t, general CRRA utility function doesn't have a balanced growth path unless it reduces to a logarithmic form.

The stochastic discount factor becomes

$$SDF_t = \beta \left(\frac{c_{t+1} - hc_t}{c_t - hc_{t-1}} \right)^{-1} = \beta \left(\frac{\gamma \tilde{c}_{t+1} - h\tilde{c}_t}{\tilde{c}_t - \frac{h}{\gamma}\tilde{c}_{t-1}} \right)^{-1}, \tag{37}$$

where $\tilde{c}_t = c_t/\gamma^t$ is the detrended variable.

The labor market and wage Calvo pricing problem follow Erceg et al. (2000), which gives us the wage Phillips curve

$$\hat{\pi}_t^w = \beta \mathbb{E}_t \hat{\pi}_{t+1}^w - \kappa_w \left(\hat{w}_t - \hat{MRS}_t \right) + \mu_{w,t}, \tag{38}$$

where $\kappa_w = \frac{(1-\theta_w)(1-\beta\theta_w)}{\theta_w(1+\nu_w\psi)}$. Because of the weak identification problem [3], ν_w is set as 3 following Smets and Wouters (2007).

$\mu_{w,t}$ is the wage markup shock following

$$\ln \mu_{w,t} = (1-\rho_w)\mu_w + \rho_w \ln \mu_{w,t-1} + \varepsilon_{w,t}, \;\; \varepsilon_{w,t} \sim \text{i.i.d. } N(0, \sigma_w^2). \tag{39}$$

The price markup shock $\mu_{p,t}$ is added in the same way—right after the Phillips curve. I assume the markup shocks follow AR(1) processes for simplicity, while Smets and Wouters (2007) uses ARMA(1, 1) processes for these two shocks.

Two added observation equations are

$$\begin{bmatrix} \Delta \ln CONS_t \\ \Delta \ln WAGE_t \end{bmatrix} = \begin{bmatrix} \bar{\gamma} \\ \bar{\gamma} \end{bmatrix} + \begin{bmatrix} \hat{c}_t - \hat{c}_{t-1} \\ \hat{w}_t - \hat{w}_{t-1} \end{bmatrix} + \Sigma_t^{meas}. \tag{40}$$

Posteriors can be found in Table 11 in Appendix C.1. Especially, both the Phillips curve and the wage Phillips curve become very flat, with κ_p estimated as 0.03 and κ_w roughly 0.02, which are much stickier than in the baseline result. A simple test that mutes the price and wage markup shocks shows that the low κ_p and κ_w partially benefit from these shocks, as they can alter price and wage exogenously, even though the endogenous part is rather sticky. The moments comparison is shown in Table 12. As explained in Section 5.1, the variance of hours worked is fitted much better.

It is clear that the investment demand shock has only a marginal effect on various macroeconomic variables, as shown in Table 7. The investment demand shock can only contribute 4.37% of the output forecast variance and 13.62% of investment, and can hardly explain other variables such as inflation. Thus I flag the fragility of the baseline result in Table 6.

Figure 10 plots the IRF of the same state variables with the IRF of baseline result. There are three points that are different from the baseline result in Figure 2. First, investment demand shock generates much less fluctuation than IST shock due to the smaller standard deviation; second, IST shock now generates inflationary result, and the positive wage impact as well; third, consumption impact under investment demand shock becomes smoother—consumption impact is only 0.2% of investment impact, whereas it is 12% in the baseline result.

Altogether, this section shows the sensitivity of the baseline result under an alternative model setting. However, its result can still be understood from the mechanisms analyzed in Section 5.

[3] ν_w, θ_w only play a role in κ_w, but κ_w can be the same under different value combinations of ν_w and θ_w. Thus the model will be unable to identify ν_w and θ_w parameters separately.

Table 7: Unconditional FEVD (%): Smets–Wouters-style model.

Observable	ε_g	ε_s	ε_R	ε_a	ε_b	ε_p	ε_w	ε_d
Output	6.15	10.31	7.86	43.58	17.21	5.00	5.52	4.37
Inflation	0.05	1.78	0.82	18.85	0.40	53.05	25.04	0.00
Interest rate	0.19	7.16	30.37	26.67	4.51	10.39	20.72	0.00
Hours	2.00	20.43	7.10	34.85	7.86	1.10	26.35	0.30
Investment	0.25	54.98	2.43	18.59	3.19	0.62	6.33	13.62
Wage	0.00	0.12	0.14	18.25	0.30	16.96	64.23	0.00
Consumption	0.25	2.36	6.48	43.49	42.01	1.92	3.49	0.00
Gov. spending	99.90	0.01	0.01	0.05	0.02	0.01	0.01	0.00

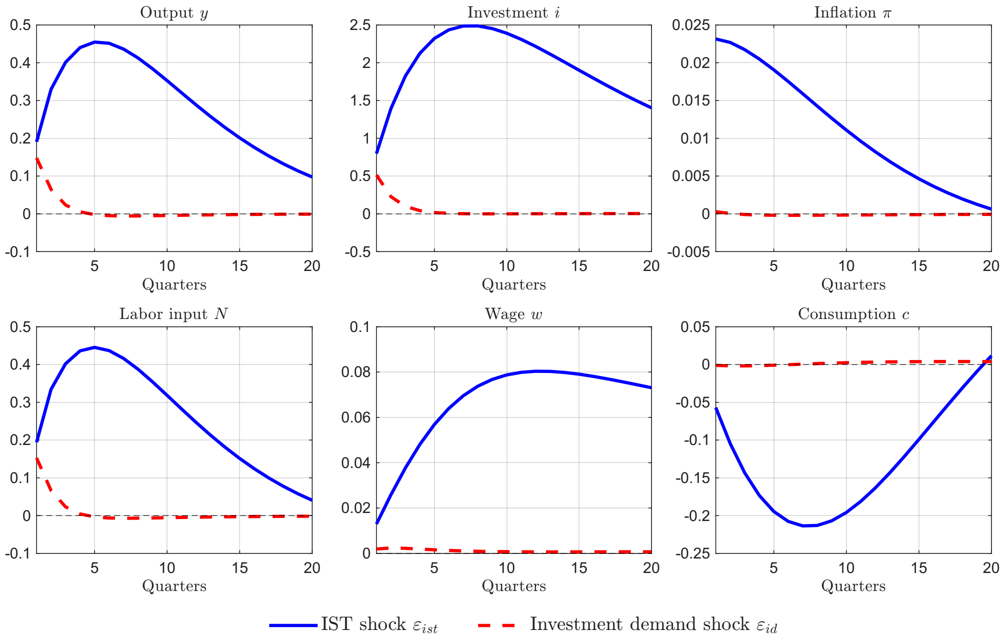

Figure 10: Impulse responses of output, investment, inflation, labor input, wage, and consumption to one-standard-deviation IST and investment demand shocks (Smets–Wouters-style model; horizon: 20 quarters).

6.2 Independent Investment Demand Shock

Although using an AR(1) process to specify shocks is standard, it is natural to think that households will realize the exogenous investment demand in the $t + 1$ period, and thus adjust their investment decision in $t + 1$ to dampen the effect of the investment demand shock. Equivalently, this is to say that the investment demand shock only has a transitory effect in period t, which motivates the exercise in this section.

All other settings are unchanged compared with the baseline, except that ρ_d is manually set as 0. To be concise, I only present the FEVD result here in Table 15, and others can be found in Table 14 and Table 15 in Appendix C.2.

Table 8: Unconditional FEVD (%): independent (i.i.d.) investment-demand model.

Observable	ε_g	ε_s	ε_R	ε_a	ε_d
Output	6.67	38.60	9.35	22.42	22.95
Inflation	0.40	52.00	12.87	34.37	0.37
Interest rate	0.37	74.14	3.74	21.66	0.09
Hours	2.84	92.22	0.12	4.53	0.29
Investment	0.25	73.06	0.20	11.31	15.18
Gov. spending	99.59	0.17	0.04	0.10	0.10

The investment demand shock contributes 22.95% of output and 15.18% of investment forecast error variance. Compared with the baseline result in Table 6, the influence of the investment demand shock also decreases as expected, since the zero persistence cuts off the future influence.

6.3 Marginal Efficiency of Investment Shock

The topic in this section is subtle—it further distinguishes the IST shock from the marginal efficiency of investment (MEI) shock. Specifically, the IST shock means the efficiency from consumption goods to investment goods, and the MEI shock means the efficiency from investment goods to capital. However, without modelling the detailed investment goods producers and capital producers, both shocks stay in the capital law of motion in their reduced form,

$$k_{t+1} = (1-\delta)k_t + \mu_{s,t}\mu_{m,t}\left[1 - S\left(\frac{i_t}{i_{t-1}}\right)\right] i_t, \tag{41}$$

and thus there will be weak identification problem within the previous data set.

This test is motivated by Justiniano et al. (2011), who claim that the IST shock will become less important if it is identified by data, and that the MEI shock becomes the one that matters. However, this raises a doubt about the estimation result—is the important thing just the residual in the capital law of motion? Although we can write multiple shocks in the capital law of motion, the equation suffers from a misspecification problem, in that we cannot be sure what the real shocks are. We can claim an IST shock, an investment demand shock, or an MEI shock in the capital law of motion, but which shocks to add depends on our assumptions. If the data-identified shock is less important, then the other part may be just the residual in the equation, and it is significant no matter what shock is named.

To alleviate the concern, the best thing to do is to identify the investment shock directly by some data. However, as this shock originates from idiosyncratic productivity shocks, the micro data that we need can hardly be incorporated into the current framework, where we only consider aggregate variables. Therefore, I can only conduct a horse-race test in this section by adding the MEI shock into the model. The result cannot conclusively establish the importance of the investment demand shock, but only offer some suggestions.

I use the relative price data to solve the weak identification problem mentioned in equation (41). As pointed out by Greenwood et al. (1997) and Fisher (2006), the relative price of investment reflects the investment specific technology level in a one-sector model [4]. Therefore, using the detrended relative price data, I assume that the detrended relative price represents the inverse of the IST level state variable. The data, the relative price of investment goods (PIRIC), is collected from FRED, and then I take $100 \ln PIRIC_t$. As the data has an obvious trend, it should be detrended to match the model assumption that the logarithm of the IST level is a stationary process. Although it is more realistic to model the logarithm of IST level as a random walk process given the data, such modeling requires recalculating the balanced growth path, steady state and log-linearized model, as another growth term is added into the model. Thus, using the detrended data is a more straightforward approach.

Table 9 reports the FEVD in the model incorporating the MEI shock and relative price data, and other results are shown in Table 16 and Table 17 in Appendix C.3. Similar to Justiniano et al. (2011), the IST shock contributes much less to the output, labor and investment forecast error compared with the baseline result in Table 6. Directly, this is because the relative price ties the scale of the IST shock, reducing the posterior mean of its standard deviation from 1.4 to 0.35. The investment demand shock still shows a strong impact on output and investment, contributing 43.16% and 48.64% of the variances of these variables.

[4] If we model the production of consumption goods and investment goods as two sectors, such as in Moura (2018), the relative price of investment goods is represented as the relative productivity, markup and marginal cost between these two sectors. Therefore, the relative price should have more complex dynamics in a more realistic model, rather than being only an exogenous stochastic process.

Table 9: Unconditional FEVD (%): MEI-shock model with relative-price data.

Observable	ε_g	ε_s	ε_R	ε_a	ε_m	ε_d
Output	4.58	0.44	10.81	23.79	17.22	43.16
Inflation	0.31	0.78	10.33	34.79	47.64	6.15
Interest rate	0.35	1.31	4.77	25.98	61.45	6.14
Hours	3.01	0.44	0.08	3.01	92.01	1.45
Investment	0.12	1.09	0.17	11.10	38.88	48.63
Gov. spending	99.65	0.00	0.04	0.09	0.06	0.16
Relative price	0.00	100.00	0.00	0.00	0.00	0.00

7 Conclusion

This paper revisits Justiniano et al. (2010), who claims that the investment specific technology is the main driving force of the business cycle, by testing an alternative hypothesis—the investment demand shock proposed by Nieri and Ragot (2025). Building a DSGE model with both the IST shock and the investment demand shock, the estimation results show that the investment demand shock can be another competing shock driving the business cycle. In the baseline result, the investment demand shock alone contributes about 40% of the forecast error of output, and about 50% of investment.

However, there are a few limitations in this paper. First, there is a lack of data to directly identify the investment demand shock, and thus the identification of the investment demand shock and the IST shock can only be based on their different effects on other state variables. Moreover, it is hard to test the argument across all combinations of parameter values, as the parameter dimension is high. Section 5.3 and 5.4 can only provide partial evidence on this argument. Second, the baseline model cannot fit the data very well, and the posteriors of some parameters take extreme values, which is discussed in Section 4.2 and 5.1. Third, the estimation results depend on the model specification. Expanding the model (Section 6.1) or using an alternative stochastic process (Section 6.2) can alter the result quite a lot.

Nevertheless, this paper highlights the insights provided by structural estimation, but also the fragility of such an approach. More macroeconomic effects of investment-side shocks can be discussed and compared, but the model specification should be selected carefully.

References

N. Ben Zeev and H. Khan. Investment-Specific News Shocks and U.S. Business Cycles. *Journal of Money, Credit and Banking*, 47(7):1443–1464, Oct. 2015. ISSN 0022-2879, 1538-4616.

L. J. Christiano, M. Eichenbaum, and C. L. Evans. Nominal Rigidities and the Dynamic Effects of a Shock to Monetary Policy. *Journal of Political Economy*, 113(1):1–45, Feb. 2005. ISSN 0022-3808.

L. J. Christiano, R. Motto, and M. Rostagno. Risk Shocks. *American Economic Review*, 104(1):27–65, Jan. 2014. ISSN 0002-8282.

C. J. Erceg, D. W. Henderson, and A. T. Levin. Optimal monetary policy with staggered wage and price contracts. *Journal of monetary Economics*, 46(2):281–313, 2000.

J. D. M. Fisher. The Dynamic Effects of Neutral and Investment-Specific Technology Shocks. *Journal of Political Economy*, 114(3):413–451, 2006. ISSN 0022-3808.

J. Greenwood, Z. Hercowitz, and G. W. Huffman. Investment, Capacity Utilization and the Real Business Cycle. *The American Economic Review*, 1988.

J. Greenwood, Z. Hercowitz, and P. Krusell. Long-Run Implications of Investment-Specific Technological Change. *The American Economic Review*, 87(3):342–362, 1997.

E. P. Herbst and F. Schorfheide. *Bayesian Estimation of DSGE Models*. Princeton University Press, 2015. ISBN 978-0-691-16108-2.

A. Justiniano, G. E. Primiceri, and A. Tambalotti. Investment shocks and business cycles. *Journal of Monetary Economics*, 57(2):132–145, Mar. 2010. ISSN 0304-3932.

A. Justiniano, G. E. Primiceri, and A. Tambalotti. Investment shocks and the relative price of investment. *Review of Economic Dynamics*, 14(1):102–121, Jan. 2011. ISSN 1094-2025.

G. Kamber, C. Smith, and C. Thoenissen. Financial frictions and the role of investment-specific technology shocks in the business cycle. *Economic Modelling*, 51:571–582, Dec. 2015. ISSN 0264-9993.

E. M. Leeper, M. Plante, and N. Traum. Dynamics of fiscal financing in the United States. *Journal of Econometrics*, 156(2):304–321, June 2010. ISSN 0304-4076.

S.-Y. Liao and B.-L. Chen. News shocks to investment-specific technology in business cycles. *European Economic Review*, 152:104363, Feb. 2023. ISSN 00142921.

A. Moura. Investment shocks, sticky prices, and the endogenous relative price of investment. *Review of Economic Dynamics*, 27:48–63, Jan. 2018. ISSN 1094-2025.

M. Nieri and X. Ragot. The Origins and Propagation of Animal Spirits Shocks. *working paper*, 2025.

F. J. Ruge-Murcia. Methods to estimate dynamic stochastic general equilibrium models. *Journal of Economic Dynamics and Control*, 31(8):2599–2636, Aug. 2007. ISSN 01651889.

S. Schmitt-Grohé and M. Uribe. What's News in Business Cycles. *Econometrica*, 80 (6):2733–2764, 2012. ISSN 1468-0262.

F. Smets and R. Wouters. Shocks and Frictions in US Business Cycles: A Bayesian DSGE Approach. *American Economic Review*, 97(3):586–606, 2007. ISSN 0002-8282.

T. Sveen and L. Weinke. New perspectives on capital, sticky prices, and the taylor principle. *Journal of Economic Theory*, 123(1):21–39, 2005.

T. Winberry. Lumpy Investment, Business Cycles, and Stimulus Policy. *American Economic Review*, 111(1):364–396, Jan. 2021. ISSN 0002-8282.

A Model Solution

This appendix derives (i) a stationary representation of the baseline model in Section 2 by detrending the real variables, (ii) the deterministic steady state, and (iii) a log-linear approximation around the steady state.

A.1 Balanced growth and detrended variables

The model features labor-augmenting growth at gross rate $\gamma > 1$. Define the deterministic trend

$$Z_t \equiv \gamma^t. \tag{42}$$

For any real quantity that shares the balanced-growth trend (output, absorption, capital, etc.), define a stationary (detrended) counterpart by dividing by Z_t:

$$\tilde{X}_t \equiv \frac{X_t}{Z_t} = \frac{X_t}{\gamma^t}, \; X_t \in \{Y_t, C_t, I_t, G_t, K_t\}. \tag{43}$$

Inflation and the gross nominal interest rate are already stationary:

$$\pi_t \equiv \frac{P_t}{P_{t-1}}, \; R_t. \tag{44}$$

Because production uses effective labor $Z_t L_t$, the real wage grows on the balanced-growth path. It is sometimes convenient to define a detrended real wage

$$\tilde{w}_t \equiv \frac{w_t}{Z_t}, \tag{45}$$

while the (gross) real return objects (e.g. R_t/π_{t+1}, the rental rate of capital services) are stationary.

Capital services and the utilization cost term also become stationary after detrending. Using $K_t^s = u_t K_{t-1}$ and $K_{t-1} = Z_{t-1}\tilde{K}_{t-1}$,

$$\tilde{K}_t^s \equiv \frac{K_t^s}{Z_t} = \frac{u_t K_{t-1}}{Z_t} = \frac{u_t}{\gamma}\tilde{K}_{t-1}. \tag{46}$$

Stationary capital accumulation. The law of motion for capital in Section 2 is

$$K_{t+1} = (1-\delta)K_t + \mu_{s,t}\left[1 - S\left(\frac{I_t}{I_{t-1}}\right)\right] I_t. \tag{47}$$

Divide by $Z_{t+1} = \gamma Z_t$ and use $I_t/I_{t-1} = \gamma\,(\tilde{I}_t/\tilde{I}_{t-1})$ to obtain the stationary representation

$$\gamma\,\tilde{K}_{t+1} = (1-\delta)\tilde{K}_t + \mu_{s,t}\left[1 - S\left(\gamma\frac{\tilde{I}_t}{\tilde{I}_{t-1}}\right)\right]\tilde{I}_t. \tag{48}$$

Under $S(x) = \frac{\phi_i}{2}(x-\gamma)^2$, the balanced-growth steady state satisfies $\tilde{I}_t/\tilde{I}_{t-1} = 1$, so that $S(\gamma) = 0$ and $S'(\gamma) = 0$.

Stationary production. For each intermediate producer, the production function is

$$Y_t(j) = \mu_{a,t}\, K_t^s(j)^\alpha\, (Z_t N_t(j))^{1-\alpha}. \tag{49}$$

Aggregating and dividing by Z_t yields the stationary aggregate production function

$$\tilde{Y}_t = \mu_{a,t}\, (\tilde{K}_t^s)^\alpha\, L_t^{1-\alpha}. \tag{50}$$

Stochastic discount factors. From equation (5),

$$\Lambda_{t,t+1} = \frac{\beta}{\gamma}\left(\frac{\tilde{C}_{t+1}}{\tilde{C}_t}\right)^{-\sigma}. \tag{51}$$

Investment optimality condition. Let $x_t \equiv I_t/I_{t-1} = \gamma\,(\tilde{I}_t/\tilde{I}_{t-1})$. The (detrended) investment FOC is given in the main text as (6).

A.2 Deterministic steady state

Consider the non-stochastic steady state of the detrended system where shocks are at their means

$$\mu_{a,t} = \mu_a,\ \mu_{s,t} = \mu_s,\ \mu_{g,t} = \mu_g, \tag{52}$$

and all stationary variables are constant: $\tilde{X}_t = \tilde{X}$, $u_t = u$, $\pi_t = \pi$, $R_t = R$, $Q_t = Q$.

Inflation and nominal interest rate. In steady state, $\pi = \pi^*$ and $Y = Y^n$ in the Taylor rule, implying a constant R. From Euler's equation,

$$1 = \frac{\beta}{\gamma}\frac{R}{\pi} \Longrightarrow R = \frac{\gamma}{\beta}\,\pi. \tag{53}$$

Investment. With $x = \gamma$ and $S(\gamma) = S'(\gamma) = 0$, the investment FOC (6) reduces to

$$1 = Q\,\mu_s \Longrightarrow Q = \mu_s^{-1}. \tag{54}$$

Under the usual normalization $\mu_s = 1$, we have $Q = 1$.

Capital accumulation. From (48) (with $S(\gamma) = 0$),

$$\tilde{I} = \big(\gamma - 1 + \delta\big)\tilde{K}\,\mu_s^{-1}. \tag{55}$$

In particular, if $\mu_s = 1$ then $\tilde{I}/\tilde{K} = \gamma - 1 + \delta$.

Return to capital. With $u = 1$, $a(1) = 0$, and constant Q, the capital Euler equation (7) implies

$$Q = \frac{\beta}{\gamma}\left(r^k + (1-\delta)Q\right) \Longrightarrow r^k = \left(\frac{\gamma}{\beta} - (1-\delta)\right)Q. \tag{56}$$

When $Q = 1$, the steady-state rental rate is $r^k = \gamma/\beta - (1-\delta)$.

Marginal cost and factor prices. With monopolistic competition and a constant elasticity of substitution ν, the desired steady-state markup is

$$\mu_p \equiv \frac{\nu}{\nu - 1}, \; mc \equiv \mu_p^{-1} = \frac{\nu - 1}{\nu}. \tag{57}$$

Cost minimization implies factor pricing conditions (real terms)

$$w = mc\,(1-\alpha)\,\frac{Y}{L}, \qquad r^k = mc\,\alpha\,\frac{Y}{K^s}. \tag{58}$$

In detrended form, $Y = Z\tilde{Y}$ and $w = Z\tilde{w}$, while L and r^k are stationary.

Wage and labor. Under preferences in effective consumption $\tilde{C}_t = C_t/Z_t$, the marginal rate of substitution implied by the household problem is

$$MRS_t \equiv -\frac{U_{L,t}}{U_{C,t}} = \chi\, L_t^{\psi}\, Z_t\, \tilde{C}_t^{\sigma}. \tag{59}$$

Hence the stationary (detrended) MRS is $\widetilde{MRS}_t \equiv MRS_t/Z_t = \chi L_t^{\psi} \tilde{C}_t^{\sigma}$. Writing the simplified sticky wage rule in stationary form,

$$\tilde{w}_t = \left(\widetilde{MRS}_t\right)^{\theta_w}\, \tilde{w}^{\,1-\theta_w}, \tag{60}$$

the steady state satisfies $\tilde{w} = \widetilde{MRS}$, i.e.

$$\tilde{w} = \chi L^{\psi} \tilde{C}^{\sigma}. \tag{61}$$

Together with goods market clearing and production, this pins down $(\tilde{C}, \tilde{I}, \tilde{Y}, \tilde{K}, L)$.

A.3 Log-linearized equilibrium conditions

For any stationary variable $\tilde{X}_t$, define the log-deviation from steady state as

$$\hat{x}_t \equiv \log\left(\frac{\tilde{X}_t}{\tilde{X}}\right). \tag{62}$$

For gross rates, define $\hat{\pi}_t \equiv \log(\pi_t/\pi)$ and $\hat{R}_t \equiv \log(R_t/R)$.

(1) Consumption Euler equation. Log-linearizing (8) yields

$$\sigma\left(\hat{c}_t - \mathbb{E}_t \hat{c}_{t+1}\right) = \hat{R}_{t+1} - \mathbb{E}_t \hat{\pi}_{t+1}. \tag{63}$$

(2) Investment. Under $S(x) = \frac{\phi_i}{2}(x-\gamma)^2$ and around $x = \gamma$, the investment first-order condition (6) log-linearizes to

$$\hat{q}_t = \hat{\mu}_{s,t} + \varphi_i\left(\hat{i}_t - \hat{i}_{t-1}\right) - \frac{\beta}{\gamma}\varphi_i\,\mathbb{E}_t\left(\hat{i}_{t+1} - \hat{i}_t\right), \tag{64}$$

where $\varphi_i \equiv \phi_i \gamma^2$ and $\hat{q}_t \equiv \log(Q_t/Q)$.

(3) Detrended capital accumulation. Since $S(\gamma) = S'(\gamma) = 0$, the adjustment cost is second order and drops out of the first-order approximation of (48):

$$\hat{k}_{t+1} = \frac{1-\delta}{\gamma}\,\hat{k}_t + \frac{\tilde{I}}{\gamma\tilde{K}}\left(\hat{\mu}_{s,t} + \hat{i}_t\right). \tag{65}$$

(4) Capital Euler equation. Let $\hat{u}_t \equiv \log(u_t)$ so that $\hat{u}_t \approx u_t - 1$ around $u = 1$. Using (7) and $\Lambda_{t,t+1}$ in (51), the log-linearized capital Euler equation can be written as

$$\hat{q}_t = -\sigma\,\mathbb{E}_t\left(\hat{c}_{t+1} - \hat{c}_t\right) + \frac{\beta}{\gamma}(1-\delta)\,\mathbb{E}_t\hat{q}_{t+1} + \frac{\beta}{\gamma}r^k\,\mathbb{E}_t\hat{r}^k_{t+1}. \tag{66}$$

where the utilization term drops out at first order under the normalization $a(1) = 0$ and $a'(1) = r^k$.

(5) Utilization optimality. Linearizing (10) around $u = 1$ yields

$$\hat{r}^k_t = \frac{a''(1)}{a'(1)}\,\hat{u}_t \equiv \frac{\psi_u}{1-\psi_u}\,\hat{u}_t. \tag{67}$$

(6) Production and factor demands. From (50),

$$\hat{y}_t = \hat{\mu}_{a,t} + \alpha\,\hat{k}^s_t + (1-\alpha)\,\hat{\ell}_t, \tag{68}$$

and since $\tilde{K}^s_t = (u_t/\gamma)\tilde{K}_{t-1}$,

$$\hat{k}^s_t = \hat{k}_{t-1} + \hat{u}_t. \tag{69}$$

Cost minimization implies

$$\hat{w}_t = \hat{mc}_t + \hat{y}_t - \hat{\ell}_t,\ \hat{r}^k_t = \hat{mc}_t + \hat{y}_t - \hat{k}^s_t. \tag{70}$$

(7) New Keynesian Phillips curve. With Calvo pricing, no indexation, and a constant desired markup, inflation obeys

$$\hat{\pi}_t = \frac{\beta}{\gamma}\,\mathbb{E}_t\hat{\pi}_{t+1} + \kappa_p\,\widehat{mc}_t, \tag{71}$$

where

$$\kappa_p \equiv \frac{(1-\theta_p)(1-\theta_p\,\beta/\gamma)}{\theta_p}. \tag{72}$$

(8) Sticky wage rule. Using $\widetilde{MRS}_t = \chi L_t^{\psi}\tilde{C}_t^{\sigma}$ and $\tilde{w}_t = (\widetilde{MRS}_t)^{\theta_w}\tilde{w}^{1-\theta_w}$, the log-linear wage equation is

$$\hat{w}_t = \theta_w\big(\sigma\hat{c}_t + \psi\hat{\ell}_t\big). \tag{73}$$

(9) Monetary policy. Log-linearizing the Taylor rule around $(\pi, Y/Y^n) = (\pi^*, 1)$ yields

$$\hat{R}_t = \rho_R\hat{R}_{t-1} + (1-\rho_R)\Big(\psi_\pi\hat{\pi}_t + \psi_y\,(\hat{y}_t - \hat{y}_t^n)\Big) + \varepsilon_{R,t}. \tag{74}$$

(10) Government spending. From $G_t = (Y_t/Y)^{-\phi_g}\mu_{g,t}$, the log-linearized rule is

$$\hat{g}_t = -\phi_g\,\hat{y}_t + \hat{\mu}_{g,t}. \tag{75}$$

(11) Resource constraint. The aggregate resource constraint $Y_t = C_t + I_t + G_t + a(u_t)K_{t-1}$ implies in detrended form

$$\tilde{Y}_t = \tilde{C}_t + \tilde{I}_t + \tilde{G}_t + \frac{1}{\gamma}a(u_t)\tilde{K}_{t-1}. \tag{76}$$

Linearizing around $a(1) = 0$ gives

$$\hat{y}_t = s_c\hat{c}_t + s_i\hat{i}_t + s_g\hat{g}_t + s_{ku}\,\hat{u}_t, \tag{77}$$

where $s_c \equiv \tilde{C}/\tilde{Y}$, $s_i \equiv \tilde{I}/\tilde{Y}$, $s_g \equiv \tilde{G}/\tilde{Y}$ and $s_{ku} \equiv (r^k\tilde{K})/(\gamma\tilde{Y})$ under the normalization $a'(1) = r^k$.

(12) Shock processes. For $x \in \{a, s, g\}$, the shock processes in Section 2 imply the stationary AR(1) laws

$$\hat{\mu}_{x,t} = \rho_x\hat{\mu}_{x,t-1} + \varepsilon_{x,t}, \tag{78}$$

with $\varepsilon_{x,t} \sim$ i.i.d. $\mathcal{N}(0, \sigma_x^2)$.

B Data

This appendix documents the construction of the observable data used in the estimation in Section 4. All series are downloaded from the Federal Reserve Economic Data (FRED) database and converted to quarterly frequency. When FRED provides a series at monthly frequency, we take the within-quarter average.

B.1 Raw series

Table 10 lists the raw FRED series used to construct the observables.

Table 10: Raw FRED series (mnemonics) used to construct the observables for estimation and robustness exercises.

Observable / input	**FRED mnemonic**
Real GDP (level)	GDPC1
CPI price level (level)	CPIAUCSL
Effective federal funds rate (level, percent)	FEDFUNDS
Civilian non-institutional population (level)	CNP16OV
Civilian employment (level)	CE16OV
Average weekly hours (index/level)	PRS85006023
GDP deflator (price level)	GDPDEF
Personal consumption expenditures (nominal)	PCEC
Private fixed investment (nominal)	FPI
Real government consumption expenditures and investment	GCEC1
Relative price of investment goods (index)	PIRIC
Compensation per hour (nominal)	COMPNFB

B.2 Transformations and scaling

The estimation uses six observables: (i) per-capita real output growth, (ii) annualized CPI inflation, (iii) the effective federal funds rate, (iv) (log) labor input, (v) per-capita real investment growth, and (vi) per-capita real government spending growth. The transformations follow the measurement conventions in Smets and Wouters (2007) and Justiniano et al. (2010).

Per-capita real output growth. Let GDP_t denote real GDP (GDPC1) and POP_t denote population (CNP16OV), both at quarterly frequency. Define real GDP per capita $GDP_t^{pc} \equiv GDP_t/POP_t$. The observable is quarterly percent growth,

$$\Delta \ln GDP_t^{pc} \equiv 100\big(\ln GDP_t^{pc} - \ln GDP_{t-1}^{pc}\big). \tag{79}$$

Annualized inflation. Let P_t denote the CPI price level (CPIAUCSL). The observable inflation rate is computed as annualized log inflation,

$$\pi_t \equiv 400\big(\ln P_t - \ln P_{t-1}\big). \tag{80}$$

Federal funds rate. Let FFR_t denote the effective federal funds rate (FEDFUNDS). This series is used in levels (percent). If downloaded at monthly frequency, it is converted to quarterly frequency by averaging within the quarter.

Labor input. Labor input combines average weekly hours (PRS85006023), employment (CE16OV), and population (CNP16OV). Let H_t denote hours, EMP_t employment, and POP_t population. The (log) labor-input observable is

$$HOURS_t \equiv 100 \ln\left(H_t \frac{EMP_t}{POP_t}\right), \tag{81}$$

and we subtract its sample mean so that $HOURS_t$ is centered at zero, consistent with the log-linear model representation.

Per-capita real investment growth. Let INV_t denote nominal private fixed investment (FPI) and DEF_t the GDP deflator (GDPDEF). The script constructs a real per-capita series

$$INV_t^{pc} \equiv \frac{INV_t}{POP_t\, DEF_t}, \tag{82}$$

and then computes quarterly percent growth,

$$\Delta \ln INV_t^{pc} \equiv 100\big(\ln INV_t^{pc} - \ln INV_{t-1}^{pc}\big). \tag{83}$$

Per-capita real government spending growth. Let GOV_t denote real government consumption expenditures and gross investment (GCEC1). The per-capita series is $GOV_t^{pc} \equiv GOV_t/POP_t$, and the observable is quarterly percent growth,

$$\Delta \ln GOV_t^{pc} \equiv 100\big(\ln GOV_t^{pc} - \ln GOV_{t-1}^{pc}\big). \tag{84}$$

Per-capita real consumption growth. For the SW-type robustness exercise in Section 6, I construct consumption growth using nominal consumption (PCEC), the GDP deflator (GDPDEF), and population (CNP16OV). Define the real per-capita consumption level

$$CONS_t \equiv \frac{PCEC_t}{POP_t\, DEF_t}, \tag{85}$$

where DEF_t denotes the GDP deflator. The observable is quarterly percent growth,

$$\Delta \ln CONS_t \equiv 100\big(\ln CONS_t - \ln CONS_{t-1}\big). \tag{86}$$

Real wage growth. Wage growth is based on nominal compensation per hour (COMPNFB) deflated by the GDP deflator (GDPDEF). Define the real wage level

$$WAGE_t \equiv \frac{COMPNFB_t}{DEF_t}, \tag{87}$$

and the observable wage growth

$$\Delta \ln WAGE_t \equiv 100\big(\ln WAGE_t - \ln WAGE_{t-1}\big). \tag{88}$$

Detrended relative price of investment. Section 6 uses the relative price of investment goods (PIRIC) as an additional observable to help identify the investment-specific technology (IST) shock. I first take logs and scale the series,

$$RP_t \equiv 100 \ln PIRIC_t, \tag{89}$$

and then remove a linear trend,

$$\widetilde{RP}_t \equiv \text{detrend}(RP_t). \tag{90}$$

C Other Results

This appendix collects the posterior summaries and the data-vs-model moment comparisons for the three robustness specifications discussed in Section 6. Each subsection's tables are confined to that subsection.

C.1 Smets–Wouters-style Model

Posterior estimates and moment fit for the Smets–Wouters-style model of Section 6.

Table 11: Posterior parameter summary for the Smets–Wouters-style model (mean, std, 5%, 95%).

Parameter	Mean	Std	5%	95%
h	0.789318	0.0331432	0.733869	0.84248
ψ	2.05906	0.507596	1.30326	2.93908
κ_p	0.0342349	0.0144691	0.0154811	0.0642415
α	0.307778	0.018153	0.277744	0.337996
θ_w	0.819954	0.0522716	0.72676	0.898419
$\bar{\ell}$	-0.377072	1.23851	-2.38813	1.67251
$\bar{\gamma}$	0.250156	0.0563519	0.154447	0.340991
$\bar{r}$	2.69376	0.537527	1.85696	3.58958

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Table 11 – continued

Parameter	Mean	Std	5%	95%
$\bar{\pi}$	4.13672	0.493083	3.37295	4.9922
ψ_π	1.65039	0.200213	1.33025	1.99108
ψ_y	0.1905	0.0926034	0.0698168	0.367066
ρ_a	0.995414	0.00223049	0.991002	0.9984
ρ_g	0.970504	0.00898614	0.956139	0.985914
ρ_s	0.847611	0.0504747	0.756227	0.922332
ρ_R	0.816709	0.030078	0.765063	0.863216
ρ_b	0.71395	0.0630923	0.604358	0.809139
ρ_p	0.37702	0.0872959	0.233339	0.519235
ρ_w	0.492563	0.111718	0.299115	0.669369
ρ_d	0.443808	0.0927867	0.276965	0.579432
σ_a	0.856932	0.0865946	0.723775	1.0095
σ_g	0.888415	0.0712977	0.777711	1.01225
σ_R	0.208141	0.0202586	0.177871	0.242672
σ_s	2.37375	0.704986	1.45391	3.76803
σ_b	0.696668	0.214898	0.410226	1.10856
σ_p	0.253546	0.0288889	0.208421	0.303183
σ_w	0.438388	0.066549	0.336518	0.556011
σ_d	0.513611	0.0879531	0.373855	0.664544
φ_i	9.82621	2.26821	6.434	13.9086
ψ_u	0.114892	0.0300344	0.0682515	0.16764
ϕ_g	0.0352812	0.0190405	0.00845613	0.0703056

Table 12: Unconditional moments, data vs. model-implied: Smets–Wouters-style model.

	Mean		Variance	
Observable	Model	Data	Model	Data
Output	0.250156	0.557311	0.721123	0.336311
Inflation	4.13672	3.08209	3.76477	2.16335
Interest rate	7.83111	6.04504	4.6454	5.00836
Hours	-0.377072	-2.97207e-14	9.83744	6.37204
Investment	0.250156	0.566225	2.79202	2.52577
Wage	0.250156	0.397143	0.720185	0.454228
Consumption	0.250156	0.655992	0.673574	0.304601
Gov. spending	0.250156	0.305833	0.823018	0.848019

Table 13: Short-run comovement of output, consumption, investment, inflation and hours: model, over the first 20 quarters. Each panel is a correlation matrix over $\{y, c, i, \pi, N\}$ (lower triangle), computed from the finite-horizon covariance $\Sigma_x^{(H)} = \sum_{h=0}^{20} \Phi^h R_x \sigma_x^2 R_x' (\Phi^h)'$, which restricts attention to the initial periods after the shock and discards the long persistent tail. The data panel is the full-sample sample correlation (linearly detrended levels), shown for reference; the data have no shock-conditional finite-horizon analog.

	y	c	i	π	N		y	c	i	π	N
Data (reference)						*Model: all shocks*					
y	1.00					y	1.00				
c	0.86	1.00				c	0.91	1.00			
i	0.42	0.64	1.00			i	0.78	0.51	1.00		
π	0.35	0.31	0.00	1.00		π	-0.15	-0.21	-0.09	1.00	
N	0.61	0.44	0.28	-0.04	1.00	N	0.56	0.37	0.60	0.00	1.00
Model: investment demand shock only						*Model: IST shock only*					
y	1.00					y	1.00				
c	-0.20	1.00				c	-0.97	1.00			
i	0.99	-0.15	1.00			i	0.96	-0.93	1.00		
π	0.46	-0.56	0.35	1.00		π	0.93	-0.87	0.81	1.00	
N	1.00	-0.21	0.99	0.48	1.00	N	0.99	-0.97	0.92	0.95	1.00

C.2 Independent Investment Demand Shock

Posterior estimates and moment fit for the model in which the investment-demand shock is i.i.d.

Table 14: Posterior parameter summary for the independent (i.i.d.) investment-demand model (mean, std, 5%, 95%).

Parameter	Mean	Std	5%	95%
σ	3.19578	0.44169	2.50627	3.95396
ψ	1.32452	0.219402	1.01932	1.71386
κ_p	0.488217	0.0746218	0.372874	0.617191
α	0.414954	0.0154983	0.390035	0.440865
θ_w	0.59064	0.0906712	0.446535	0.744111
$\bar{\ell}$	0.760381	1.40463	-1.58666	3.06506
$\bar{\gamma}$	0.355719	0.0217611	0.318886	0.39047
$\bar{r}$	2.24793	0.423956	1.5245	2.93154
$\bar{\pi}$	4.41421	0.689487	3.4686	5.55323
ψ_π	1.80523	0.195855	1.47337	2.12852
ψ_y	0.407772	0.227664	0.130916	0.842778
ρ_a	0.894964	0.0333533	0.841777	0.957558
ρ_g	0.970219	0.0127162	0.947788	0.988996
ρ_s	0.923814	0.0259531	0.878598	0.963433
ρ_R	0.766888	0.0404594	0.701066	0.833808

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Table 14 – continued

Parameter	Mean	Std	5%	95%
σ_a	0.474856	0.0830591	0.327078	0.60536
σ_g	0.903155	0.0750312	0.786836	1.02981
σ_R	0.244644	0.0270224	0.202964	0.291997
σ_s	3.05003	0.60494	2.10212	4.09423
σ_d	0.549554	0.126359	0.383514	0.788584
φ_i	6.39966	1.83437	3.72635	9.71367
ψ_u	0.139581	0.0436726	0.0689582	0.21288
ϕ_g	0.0779049	0.0515627	0.014074	0.177194

Table 15: Unconditional moments, data vs. model-implied: independent (i.i.d.) investment-demand model.

	Mean		Variance	
Observable	Model	Data	Model	Data
Output	0.355719	0.557311	0.599824	0.336311
Inflation	4.41421	3.08209	6.18871	2.16335
Interest rate	8.08501	6.04504	10.8	5.00836
Hours	0.760381	-2.97207e-14	16.1817	6.37204
Investment	0.355719	0.566225	4.12066	2.52577
Gov. spending	0.355719	0.305833	0.837917	0.848019

C.3 Marginal Efficiency of Investment Shock

Posterior estimates and moment fit for the model that adds the MEI shock and the relative-price observable.

Table 16: Posterior parameter summary for the MEI-shock model with relative-price data (mean, std, 5%, 95%).

Parameter	Mean	Std	5%	95%
σ	2.72532	0.410343	2.09831	3.43364
ψ	1.87221	0.292736	1.41723	2.39334
κ_p	0.399372	0.0681425	0.294782	0.517415
α	0.40241	0.0179806	0.372085	0.431748
θ_w	0.461886	0.0952085	0.317097	0.626127
$\bar{\ell}$	1.02235	1.77618	-1.92085	3.86915
$\bar{\gamma}$	0.330592	0.0266304	0.284192	0.372339
$\bar{r}$	1.72302	0.445638	0.95412	2.43557
$\bar{\pi}$	4.32199	0.899677	3.10971	6.00596
ψ_π	1.80683	0.195192	1.50385	2.14188
ψ_y	0.28156	0.122336	0.107085	0.508271
ρ_a	0.940061	0.0277589	0.891146	0.981549
ρ_g	0.989801	0.00376522	0.982824	0.995217

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Table 16 – continued

Parameter	Mean	Std	5%	95%
ρ_s	0.924708	0.0239768	0.883683	0.961365
ρ_R	0.775321	0.0384891	0.707226	0.834059
σ_a	0.440538	0.0640171	0.343732	0.552686
σ_g	0.891235	0.0702004	0.781428	1.01108
σ_R	0.239654	0.0271635	0.199584	0.289978
σ_s	0.420755	0.0419203	0.354669	0.492266
ρ_d	0.869291	0.0414575	0.795151	0.931259
σ_d	1.36431	0.175211	1.10039	1.68022
φ_i	8.56207	2.02662	5.55579	12.1727
ψ_u	0.0836911	0.0299383	0.0393241	0.135858
ϕ_g	0.065442	0.0434625	0.0119007	0.151993
ρ_m	0.989372	0.00553808	0.978574	0.996856
σ_m	1.52804	0.344867	0.981752	2.12499

Table 17: Unconditional moments, data vs. model-implied: MEI-shock model with relative-price data.

	Mean		Variance	
Observable	Model	Data	Model	Data
Output	0.330592	0.557311	0.687462	0.336311
Inflation	4.32199	3.08209	6.63809	2.16335
Interest rate	7.36738	6.04504	10.3437	5.00836
Hours	1.02235	-2.97207e-14	39.7892	6.37204
Investment	0.330592	0.566225	4.19753	2.52577
Gov. spending	0.330592	0.305833	0.814836	0.848019
Relative price	0	-0.00949494	1.30848	2.19818